# Model for "Mn3Sn"

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## General Condition

- Basis type: `lg`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [$A_{1g}$, $A_{2g}$, $B_{1g}$, $B_{2g}$, $E_{1g}$, $E_{2g}$, $A_{1u}$, $A_{2u}$, $B_{1u}$, $B_{2u}$, $E_{1u}$, $E_{2u}$]
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [$A_{1g}$, $A_{2g}$, $B_{1g}$, $B_{2g}$, $E_{1g}$, $E_{2g}$, $A_{1u}$, $A_{2u}$, $B_{1u}$, $B_{2u}$, $E_{1u}$, $E_{2u}$]
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [$A_{1g}$, $A_{2g}$, $B_{1g}$, $B_{2g}$, $E_{1g}$, $E_{2g}$, $A_{1u}$, $A_{2u}$, $B_{1u}$, $B_{2u}$, $E_{1u}$, $E_{2u}$]
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [$A_{1g}$, $A_{2g}$, $B_{1g}$, $B_{2g}$, $E_{1g}$, $E_{2g}$, $A_{1u}$, $A_{2u}$, $B_{1u}$, $B_{2u}$, $E_{1u}$, $E_{2u}$]
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

## Group and Unit Cell

- Group: SG No. 194 $D_{6\mathrm{h}}^4$ $P6_3/mmc$ [ hexagonal ]
- Associated point group: PG No. 194 $D_{6\mathrm{h}}$ $6/mmm$ [ hexagonal ]
- Unit cell:
  $a =$ `1.00000`, $b =$ `1.00000`, $c =$ `1.00000`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `120.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 1.00000, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[-0.50000, 0.86603, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000, 1.00000]`

## Symmetry Operation

Table 1: Symmetry operation

| # | SO | # | SO | # | SO | # | SO | # | SO |
|---|---|---|---|---|---|---|---|---|---|
| 1 | $\{1\|0\}$ | 2 | $\{3^{+}_{001}\|0\}$ | 3 | $\{3^{-}_{001}\|0\}$ | 4 | $\{2_{001}\|00\frac{1}{2}\}$ | 5 | $\{6^{-}_{001}\|00\frac{1}{2}\}$ |
| 6 | $\{6^{+}_{001}\|00\frac{1}{2}\}$ | 7 | $\{2_{110}\|0\}$ | 8 | $\{2_{100}\|0\}$ | 9 | $\{2_{010}\|0\}$ | 10 | $\{2_{1-10}\|00\frac{1}{2}\}$ |
| 11 | $\{2_{120}\|00\frac{1}{2}\}$ | 12 | $\{2_{210}\|00\frac{1}{2}\}$ | 13 | $\{-1\|0\}$ | 14 | $\{-3^{+}_{001}\|0\}$ | 15 | $\{-3^{-}_{001}\|0\}$ |
| 16 | $\{\mathrm{m}_{001}\|00\frac{1}{2}\}$ | 17 | $\{-6^{-}_{001}\|00\frac{1}{2}\}$ | 18 | $\{-6^{+}_{001}\|00\frac{1}{2}\}$ | 19 | $\{\mathrm{m}_{110}\|0\}$ | 20 | $\{\mathrm{m}_{100}\|0\}$ |
| 21 | $\{\mathrm{m}_{010}\|0\}$ | 22 | $\{\mathrm{m}_{1-10}\|00\frac{1}{2}\}$ | 23 | $\{\mathrm{m}_{120}\|00\frac{1}{2}\}$ | 24 | $\{\mathrm{m}_{210}\|00\frac{1}{2}\}$ | | |

## Harmonics

Table 2: Harmonics

| # | symbol | irrep. | rank | X | multiplicity | component | symmetry |
|---|---|---|---|---|---|---|---|
| 1 | $\mathbb{Q}_0(A_{1g})$ | $A_{1g}$ | 0 | $Q$, $T$ | - | - | 1 |
| 2 | $\mathbb{Q}_2(A_{1g})$ | $A_{1g}$ | 2 | $Q$, $T$ | - | - | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| 3 | $\mathbb{Q}_4(A_{1g})$ | $A_{1g}$ | 4 | $Q$, $T$ | - | - | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| 4 | $\mathbb{Q}_6(A_{1g},2)$ | $A_{1g}$ | 6 | $Q$, $T$ | 2 | - | $\frac{\sqrt{462}(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)}{32}$ |
| 5 | $\mathbb{G}_0(A_{1u})$ | $A_{1u}$ | 0 | $G$, $M$ | - | - | 1 |
| 6 | $\mathbb{G}_2(A_{1u})$ | $A_{1u}$ | 2 | $G$, $M$ | - | - | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 2

| # | symbol | irrep. | rank | X | multiplicity | component | symmetry |
|---|---|---|---|---|---|---|---|
| 7 | $\mathbb{G}_4(A_{1u})$ | $A_{1u}$ | 4 | $G,\ M$ | - | - | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| 8 | $\mathbb{G}_6(A_{1u},2)$ | $A_{1u}$ | 6 | $G,\ M$ | 2 | - | $\frac{\sqrt{462}(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)}{32}$ |
| 9 | $\mathbb{Q}_7(A_{1u})$ | $A_{1u}$ | 7 | $Q,\ T$ | - | - | $\frac{\sqrt{6006}xyz\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$ |
| 10 | $\mathbb{G}_1(A_{2g})$ | $A_{2g}$ | 1 | $G,\ M$ | - | - | $z$ |
| 11 | $\mathbb{G}_3(A_{2g})$ | $A_{2g}$ | 3 | $G,\ M$ | - | - | $-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$ |
| 12 | $\mathbb{G}_5(A_{2g})$ | $A_{2g}$ | 5 | $G,\ M$ | - | - | $\frac{z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{8}$ |
| 13 | $\mathbb{Q}_6(A_{2g})$ | $A_{2g}$ | 6 | $Q,\ T$ | - | - | $\frac{\sqrt{462}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$ |
| 14 | $\mathbb{Q}_1(A_{2u})$ | $A_{2u}$ | 1 | $Q,\ T$ | - | - | $z$ |
| 15 | $\mathbb{Q}_3(A_{2u})$ | $A_{2u}$ | 3 | $Q,\ T$ | - | - | $-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$ |
| 16 | $\mathbb{G}_6(A_{2u})$ | $A_{2u}$ | 6 | $G,\ M$ | - | - | $\frac{\sqrt{462}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$ |
| 17 | $\mathbb{Q}_7(A_{2u},2)$ | $A_{2u}$ | 7 | $Q,\ T$ | 2 | - | $\frac{\sqrt{6006}z(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)}{32}$ |
| 18 | $\mathbb{G}_3(B_{1g})$ | $B_{1g}$ | 3 | $G,\ M$ | - | - | $\frac{\sqrt{10}y\left(3x^2-y^2\right)}{4}$ |
| 19 | $\mathbb{Q}_4(B_{1g})$ | $B_{1g}$ | 4 | $Q,\ T$ | - | - | $\frac{\sqrt{70}xz\left(x^2-3y^2\right)}{4}$ |
| 20 | $\mathbb{Q}_6(B_{1g})$ | $B_{1g}$ | 6 | $Q,\ T$ | - | - | $-\frac{\sqrt{210}xz\left(x^2-3y^2\right)\left(3x^2+3y^2-8z^2\right)}{16}$ |
| 21 | $\mathbb{Q}_3(B_{1u})$ | $B_{1u}$ | 3 | $Q,\ T$ | - | - | $\frac{\sqrt{10}y\left(3x^2-y^2\right)}{4}$ |
| 22 | $\mathbb{G}_4(B_{1u})$ | $B_{1u}$ | 4 | $G,\ M$ | - | - | $\frac{\sqrt{70}xz\left(x^2-3y^2\right)}{4}$ |
| 23 | $\mathbb{Q}_5(B_{1u})$ | $B_{1u}$ | 5 | $Q,\ T$ | - | - | $-\frac{\sqrt{70}y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{16}$ |
| 24 | $\mathbb{Q}_9(B_{1u},1)$ | $B_{1u}$ | 9 | $Q,\ T$ | 1 | - | $\frac{\sqrt{24310}y\left(3x^2-y^2\right)\left(3x^6-27x^4y^2+33x^2y^4-y^6\right)}{256}$ |
| 25 | $\mathbb{G}_3(B_{2g})$ | $B_{2g}$ | 3 | $G,\ M$ | - | - | $\frac{\sqrt{10}x\left(x^2-3y^2\right)}{4}$ |
| 26 | $\mathbb{Q}_4(B_{2g})$ | $B_{2g}$ | 4 | $Q,\ T$ | - | - | $\frac{\sqrt{70}yz\left(3x^2-y^2\right)}{4}$ |
| 27 | $\mathbb{Q}_6(B_{2g})$ | $B_{2g}$ | 6 | $Q,\ T$ | - | - | $-\frac{\sqrt{210}yz\left(3x^2-y^2\right)\left(3x^2+3y^2-8z^2\right)}{16}$ |

*continued ...*

Table 2

| # | symbol | irrep. | rank | X | multiplicity | component | symmetry |
|---|---|---|---|---|---|---|---|
| 28 | $\mathbb{Q}_3(B_{2u})$ | $B_{2u}$ | 3 | $Q$, $T$ | - | - | $\frac{\sqrt{10}x\left(x^2-3y^2\right)}{4}$ |
| 29 | $\mathbb{G}_4(B_{2u})$ | $B_{2u}$ | 4 | $G$, $M$ | - | - | $\frac{\sqrt{70}yz\left(3x^2-y^2\right)}{4}$ |
| 30 | $\mathbb{Q}_5(B_{2u})$ | $B_{2u}$ | 5 | $Q$, $T$ | - | - | $-\frac{\sqrt{70}x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{16}$ |
| 31 | $\mathbb{Q}_9(B_{2u},1)$ | $B_{2u}$ | 9 | $Q$, $T$ | 1 | - | $\frac{\sqrt{24310}x\left(x^2-3y^2\right)\left(x^6-33x^4y^2+27x^2y^4-3y^6\right)}{256}$ |
| 32 | $\mathbb{G}_{1,1}(E_{1g})$ | $E_{1g}$ | 1 | $G$, $M$ | - | 1 | $x$ |
| 33 | $\mathbb{G}_{1,2}(E_{1g})$ | | | | | 2 | $y$ |
| 34 | $\mathbb{Q}_{2,1}(E_{1g})$ | $E_{1g}$ | 2 | $Q$, $T$ | - | 1 | $\sqrt{3}yz$ |
| 35 | $\mathbb{Q}_{2,2}(E_{1g})$ | | | | | 2 | $-\sqrt{3}xz$ |
| 36 | $\mathbb{G}_{3,1}(E_{1g})$ | $E_{1g}$ | 3 | $G$, $M$ | - | 1 | $-\frac{\sqrt{6}x\left(x^2+y^2-4z^2\right)}{4}$ |
| 37 | $\mathbb{G}_{3,2}(E_{1g})$ | | | | | 2 | $-\frac{\sqrt{6}y\left(x^2+y^2-4z^2\right)}{4}$ |
| 38 | $\mathbb{Q}_{4,1}(E_{1g})$ | $E_{1g}$ | 4 | $Q$, $T$ | - | 1 | $-\frac{\sqrt{10}yz\left(3x^2+3y^2-4z^2\right)}{4}$ |
| 39 | $\mathbb{Q}_{4,2}(E_{1g})$ | | | | | 2 | $\frac{\sqrt{10}xz\left(3x^2+3y^2-4z^2\right)}{4}$ |
| 40 | $\mathbb{G}_{5,1}(E_{1g},1)$ | $E_{1g}$ | 5 | $G$, $M$ | 1 | 1 | $\frac{3\sqrt{14}x\left(x^4-10x^2y^2+5y^4\right)}{16}$ |
| 41 | $\mathbb{G}_{5,2}(E_{1g},1)$ | | | | | 2 | $-\frac{3\sqrt{14}y\left(5x^4-10x^2y^2+y^4\right)}{16}$ |
| 42 | $\mathbb{Q}_{6,1}(E_{1g},1)$ | $E_{1g}$ | 6 | $Q$, $T$ | 1 | 1 | $\frac{3\sqrt{154}yz\left(5x^4-10x^2y^2+y^4\right)}{16}$ |
| 43 | $\mathbb{Q}_{6,2}(E_{1g},1)$ | | | | | 2 | $\frac{3\sqrt{154}xz\left(x^4-10x^2y^2+5y^4\right)}{16}$ |
| 44 | $\mathbb{Q}_{6,1}(E_{1g},2)$ | $E_{1g}$ | 6 | $Q$, $T$ | 2 | 1 | $\frac{\sqrt{21}yz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{8}$ |
| 45 | $\mathbb{Q}_{6,2}(E_{1g},2)$ | | | | | 2 | $-\frac{\sqrt{21}xz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{8}$ |
| 46 | $\mathbb{Q}_{8,1}(E_{1g},1)$ | $E_{1g}$ | 8 | $Q$, $T$ | 1 | 1 | $\frac{3\sqrt{715}yz\left(7x^6-35x^4y^2+21x^2y^4-y^6\right)}{32}$ |
| 47 | $\mathbb{Q}_{8,2}(E_{1g},1)$ | | | | | 2 | $-\frac{3\sqrt{715}xz\left(x^6-21x^4y^2+35x^2y^4-7y^6\right)}{32}$ |
| 48 | $\mathbb{Q}_{1,1}(E_{1u})$ | $E_{1u}$ | 1 | $Q$, $T$ | - | 1 | $x$ |

*continued ...*

Table 2

| # | symbol | irrep. | rank | X | multiplicity | component | symmetry |
|---|---|---|---|---|---|---|---|
| 49 | $\mathbb{Q}_{1,2}(E_{1u})$ | | | | | 2 | $y$ |
| 50 | $\mathbb{G}_{2,1}(E_{1u})$ | $E_{1u}$ | 2 | $G$, $M$ | - | 1 | $\sqrt{3}yz$ |
| 51 | $\mathbb{G}_{2,2}(E_{1u})$ | | | | | 2 | $-\sqrt{3}xz$ |
| 52 | $\mathbb{Q}_{3,1}(E_{1u})$ | $E_{1u}$ | 3 | $Q$, $T$ | - | 1 | $-\frac{\sqrt{6}x\left(x^2+y^2-4z^2\right)}{4}$ |
| 53 | $\mathbb{Q}_{3,2}(E_{1u})$ | | | | | 2 | $-\frac{\sqrt{6}y\left(x^2+y^2-4z^2\right)}{4}$ |
| 54 | $\mathbb{G}_{4,1}(E_{1u})$ | $E_{1u}$ | 4 | $G$, $M$ | - | 1 | $-\frac{\sqrt{10}yz\left(3x^2+3y^2-4z^2\right)}{4}$ |
| 55 | $\mathbb{G}_{4,2}(E_{1u})$ | | | | | 2 | $\frac{\sqrt{10}xz\left(3x^2+3y^2-4z^2\right)}{4}$ |
| 56 | $\mathbb{Q}_{5,1}(E_{1u},1)$ | $E_{1u}$ | 5 | $Q$, $T$ | 1 | 1 | $\frac{3\sqrt{14}x\left(x^4-10x^2y^2+5y^4\right)}{16}$ |
| 57 | $\mathbb{Q}_{5,2}(E_{1u},1)$ | | | | | 2 | $-\frac{3\sqrt{14}y\left(5x^4-10x^2y^2+y^4\right)}{16}$ |
| 58 | $\mathbb{Q}_{5,1}(E_{1u},2)$ | $E_{1u}$ | 5 | $Q$, $T$ | 2 | 1 | $\frac{\sqrt{15}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{8}$ |
| 59 | $\mathbb{Q}_{5,2}(E_{1u},2)$ | | | | | 2 | $\frac{\sqrt{15}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{8}$ |
| 60 | $\mathbb{Q}_{7,1}(E_{1u},1)$ | $E_{1u}$ | 7 | $Q$, $T$ | 1 | 1 | $\frac{\sqrt{429}x\left(x^6-21x^4y^2+35x^2y^4-7y^6\right)}{32}$ |
| 61 | $\mathbb{Q}_{7,2}(E_{1u},1)$ | | | | | 2 | $\frac{\sqrt{429}y\left(7x^6-35x^4y^2+21x^2y^4-y^6\right)}{32}$ |
| 62 | $\mathbb{Q}_{2,1}(E_{2g})$ | $E_{2g}$ | 2 | $Q$, $T$ | - | 1 | $\frac{\sqrt{3}(x-y)(x+y)}{2}$ |
| 63 | $\mathbb{Q}_{2,2}(E_{2g})$ | | | | | 2 | $-\sqrt{3}xy$ |
| 64 | $\mathbb{G}_{3,1}(E_{2g})$ | $E_{2g}$ | 3 | $G$, $M$ | - | 1 | $\sqrt{15}xyz$ |
| 65 | $\mathbb{G}_{3,2}(E_{2g})$ | | | | | 2 | $\frac{\sqrt{15}z(x-y)(x+y)}{2}$ |
| 66 | $\mathbb{Q}_{4,1}(E_{2g},1)$ | $E_{2g}$ | 4 | $Q$, $T$ | 1 | 1 | $\frac{\sqrt{35}\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$ |
| 67 | $\mathbb{Q}_{4,2}(E_{2g},1)$ | | | | | 2 | $\frac{\sqrt{35}xy(x-y)(x+y)}{2}$ |
| 68 | $\mathbb{Q}_{4,1}(E_{2g},2)$ | $E_{2g}$ | 4 | $Q$, $T$ | 2 | 1 | $-\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$ |
| 69 | $\mathbb{Q}_{4,2}(E_{2g},2)$ | | | | | 2 | $\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$ |

*continued ...*

Table 2

| # | symbol | irrep. | rank | X | multiplicity | component | symmetry |
|---|---|---|---|---|---|---|---|
| 70 | $\mathbb{Q}_{6,1}(E_{2g},1)$ | $E_{2g}$ | 6 | $Q$, $T$ | 1 | 1 | $-\frac{3\sqrt{7}\left(x^2+y^2-10z^2\right)\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{16}$ |
| 71 | $\mathbb{Q}_{6,2}(E_{2g},1)$ | | | | | 2 | $-\frac{3\sqrt{7}xy(x-y)(x+y)\left(x^2+y^2-10z^2\right)}{4}$ |
| 72 | $\mathbb{Q}_{6,1}(E_{2g},2)$ | $E_{2g}$ | 6 | $Q$, $T$ | 2 | 1 | $\frac{\sqrt{210}(x-y)(x+y)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{32}$ |
| 73 | $\mathbb{Q}_{6,2}(E_{2g},2)$ | | | | | 2 | $-\frac{\sqrt{210}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{16}$ |
| 74 | $\mathbb{Q}_{8,1}(E_{2g},1)$ | $E_{2g}$ | 8 | $Q$, $T$ | 1 | 1 | $\frac{3\sqrt{715}\left(x^4-4x^3y-6x^2y^2+4xy^3+y^4\right)\left(x^4+4x^3y-6x^2y^2-4xy^3+y^4\right)}{128}$ |
| 75 | $\mathbb{Q}_{8,2}(E_{2g},1)$ | | | | | 2 | $-\frac{3\sqrt{715}xy(x-y)(x+y)\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{16}$ |
| 76 | $\mathbb{G}_{2,1}(E_{2u})$ | $E_{2u}$ | 2 | $G$, $M$ | - | 1 | $\frac{\sqrt{3}(x-y)(x+y)}{2}$ |
| 77 | $\mathbb{G}_{2,2}(E_{2u})$ | | | | | 2 | $-\sqrt{3}xy$ |
| 78 | $\mathbb{Q}_{3,1}(E_{2u})$ | $E_{2u}$ | 3 | $Q$, $T$ | - | 1 | $\sqrt{15}xyz$ |
| 79 | $\mathbb{Q}_{3,2}(E_{2u})$ | | | | | 2 | $\frac{\sqrt{15}z(x-y)(x+y)}{2}$ |
| 80 | $\mathbb{G}_{4,1}(E_{2u},1)$ | $E_{2u}$ | 4 | $G$, $M$ | 1 | 1 | $\frac{\sqrt{35}\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$ |
| 81 | $\mathbb{G}_{4,2}(E_{2u},1)$ | | | | | 2 | $\frac{\sqrt{35}xy(x-y)(x+y)}{2}$ |
| 82 | $\mathbb{G}_{4,1}(E_{2u},2)$ | $E_{2u}$ | 4 | $G$, $M$ | 2 | 1 | $-\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$ |
| 83 | $\mathbb{G}_{4,2}(E_{2u},2)$ | | | | | 2 | $\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$ |
| 84 | $\mathbb{Q}_{5,1}(E_{2u},1)$ | $E_{2u}$ | 5 | $Q$, $T$ | 1 | 1 | $-\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$ |
| 85 | $\mathbb{Q}_{5,2}(E_{2u},1)$ | | | | | 2 | $\frac{3\sqrt{35}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$ |
| 86 | $\mathbb{Q}_{5,1}(E_{2u},2)$ | $E_{2u}$ | 5 | $Q$, $T$ | 2 | 1 | $-\frac{\sqrt{105}xyz\left(x^2+y^2-2z^2\right)}{2}$ |
| 87 | $\mathbb{Q}_{5,2}(E_{2u},2)$ | | | | | 2 | $-\frac{\sqrt{105}z(x-y)(x+y)\left(x^2+y^2-2z^2\right)}{4}$ |
| 88 | $\mathbb{Q}_{9,1}(E_{2u},1)$ | $E_{2u}$ | 9 | $Q$, $T$ | 1 | 1 | $\frac{3\sqrt{12155}xyz(x-y)(x+y)\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{16}$ |
| 89 | $\mathbb{Q}_{9,2}(E_{2u},1)$ | | | | | 2 | $\frac{3\sqrt{12155}z\left(x^4-4x^3y-6x^2y^2+4xy^3+y^4\right)\left(x^4+4x^3y-6x^2y^2-4xy^3+y^4\right)}{128}$ |

Table 3: dimension = 40

| # | orbital@atom(SL) | # | orbital@atom(SL) | # | orbital@atom(SL) | # | orbital@atom(SL) | # | orbital@atom(SL) |
|---|---|---|---|---|---|---|---|---|---|
| 0 | $\|d_v\rangle$ @Mn(1) | 1 | $\|d_{xy}\rangle$ @Mn(1) | 2 | $\|d_{xz}\rangle$ @Mn(1) | 3 | $\|d_{yz}\rangle$ @Mn(1) | 4 | $\|d_u\rangle$ @Mn(1) |
| 5 | $\|d_v\rangle$ @Mn(2) | 6 | $\|d_{xy}\rangle$ @Mn(2) | 7 | $\|d_{xz}\rangle$ @Mn(2) | 8 | $\|d_{yz}\rangle$ @Mn(2) | 9 | $\|d_u\rangle$ @Mn(2) |
| 10 | $\|d_v\rangle$ @Mn(3) | 11 | $\|d_{xy}\rangle$ @Mn(3) | 12 | $\|d_{xz}\rangle$ @Mn(3) | 13 | $\|d_{yz}\rangle$ @Mn(3) | 14 | $\|d_u\rangle$ @Mn(3) |
| 15 | $\|d_v\rangle$ @Mn(4) | 16 | $\|d_{xy}\rangle$ @Mn(4) | 17 | $\|d_{xz}\rangle$ @Mn(4) | 18 | $\|d_{yz}\rangle$ @Mn(4) | 19 | $\|d_u\rangle$ @Mn(4) |
| 20 | $\|d_v\rangle$ @Mn(5) | 21 | $\|d_{xy}\rangle$ @Mn(5) | 22 | $\|d_{xz}\rangle$ @Mn(5) | 23 | $\|d_{yz}\rangle$ @Mn(5) | 24 | $\|d_u\rangle$ @Mn(5) |
| 25 | $\|d_v\rangle$ @Mn(6) | 26 | $\|d_{xy}\rangle$ @Mn(6) | 27 | $\|d_{xz}\rangle$ @Mn(6) | 28 | $\|d_{yz}\rangle$ @Mn(6) | 29 | $\|d_u\rangle$ @Mn(6) |
| 30 | $\|d_v\rangle$ @Sn(1) | 31 | $\|d_{xy}\rangle$ @Sn(1) | 32 | $\|d_{xz}\rangle$ @Sn(1) | 33 | $\|d_{yz}\rangle$ @Sn(1) | 34 | $\|d_u\rangle$ @Sn(1) |
| 35 | $\|d_v\rangle$ @Sn(2) | 36 | $\|d_{xy}\rangle$ @Sn(2) | 37 | $\|d_{xz}\rangle$ @Sn(2) | 38 | $\|d_{yz}\rangle$ @Sn(2) | 39 | $\|d_u\rangle$ @Sn(2) |

Table 4: Atomic basis (orbital part only)

| orbital | definition |
|---|---|
| $\|d_v\rangle$ | $\frac{\sqrt{3}(x^2-y^2)}{2}$ |
| $\|d_{xy}\rangle$ | $\sqrt{3}xy$ |
| $\|d_{xz}\rangle$ | $\sqrt{3}xz$ |
| $\|d_{yz}\rangle$ | $\sqrt{3}yz$ |
| $\|d_u\rangle$ | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

## SAMB: 720 (all 720)

- `Mn` : '`Mn`' site-cluster
  * bra: $\langle d_v|$ , $\langle d_{xy}|$ , $\langle d_{xz}|$ , $\langle d_{yz}|$ , $\langle d_u|$
  * ket: $|d_v\rangle$ , $|d_{xy}\rangle$ , $|d_{xz}\rangle$ , $|d_{yz}\rangle$ , $|d_u\rangle$
  * wyckoff: `6h`

$\boxed{\text{z1}}$ $\mathbb{Q}_0^{(c)}(A_{1g},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z2}}$ $\mathbb{Q}_0^{(c)}(A_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z3}}$ $\mathbb{Q}_2^{(c)}(A_{1g},a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z4}}$ $\mathbb{Q}_2^{(c)}(A_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z5}}$ $\mathbb{Q}_4^{(c)}(A_{1g}) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z6}}$ $\mathbb{Q}_6^{(c)}(A_{1g},2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z41}}$ $\mathbb{Q}_7^{(c)}(A_{1u}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$

$\boxed{\text{z42}}$ $\mathbb{G}_2^{(c)}(A_{1u}) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z43}}$ $\mathbb{G}_4^{(c)}(A_{1u}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z67}}$ $\mathbb{Q}_6^{(c)}(A_{2g}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z68}}$ $\mathbb{G}_1^{(c)}(A_{2g}) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z69}}$ $\mathbb{G}_3^{(c)}(A_{2g}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z95}}$ $\mathbb{Q}_1^{(c)}(A_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z96}}$ $\mathbb{Q}_1^{(c)}(A_{2u},b) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_3^{(s)}(B_{1u})$

$\boxed{\text{z97}}$ $\mathbb{Q}_3^{(c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z126}}$ $\mathbb{Q}_4^{(c)}(B_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z127}}$ $\mathbb{Q}_4^{(c)}(B_{1g},b) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z128}}$ $\mathbb{Q}_4^{(c)}(B_{1g},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z129}}$ $\mathbb{Q}_3^{(c)}(B_{1u},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$

$\boxed{\text{z130}}$ $\mathbb{Q}_3^{(c)}(B_{1u},b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$

$\boxed{\text{z131}}$ $\mathbb{Q}_3^{(c)}(B_{1u},c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z132}}$ $\mathbb{Q}_3^{(c)}(B_{1u},d) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$

$\boxed{\text{z133}}$ $\mathbb{Q}_3^{(c)}(B_{1u},e) = -\frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{58}$

$\boxed{\text{z134}}$ $\mathbb{Q}_5^{(c)}(B_{1u}) = -\frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{58} - \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{29}$

$\boxed{\text{z135}}$ $\mathbb{Q}_4^{(c)}(B_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z136}}$ $\mathbb{Q}_4^{(c)}(B_{2g},b) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z137}}$ $\mathbb{Q}_4^{(c)}(B_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z244}}$ $\mathbb{Q}_{3}^{(c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z245}}$ $\mathbb{Q}_{3}^{(c)}(B_{2u},b) = \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{58}$

$\boxed{\text{z246}}$ $\mathbb{Q}_{5}^{(c)}(B_{2u}) = \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{29}$

$\boxed{\text{z247}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z248}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z249}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z250}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z251}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},c) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8}$

$\boxed{\text{z252}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8}$

$\boxed{\text{z253}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z254}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z255}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{1g},b) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8}$

$\boxed{\text{z256}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{1g},b) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{8}$

$\boxed{\text{z257}}$ $\mathbb{Q}_{6,1}^{(c)}(E_{1g},1) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$

$\boxed{\text{z258}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_{1g},1) = \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} + \frac{\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$$

$\boxed{\text{z259}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2}$$

$\boxed{\text{z260}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2}$$

$\boxed{\text{z261}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},b) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$\boxed{\text{z376}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$\boxed{\text{z377}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},c) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{14}$$

$\boxed{\text{z378}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},c) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{14}$$

$\boxed{\text{z404}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},d) = \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{58}$$

$\boxed{\text{z405}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},d) = -\frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{58}$$

$\boxed{\text{z406}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},a) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{14} + \frac{\sqrt{21}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{7}$$

$\boxed{\text{z430}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},a) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{14} + \frac{\sqrt{21}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{7}$$

$\boxed{\text{z431}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},b) = \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{58} - \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{29}$$

$\boxed{\text{z432}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},b) = -\frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{29}$$

$\boxed{\text{z433}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},c) = \frac{\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{14} - \frac{\sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{7}$$

$\boxed{\text{z434}}$ $\mathbb{Q}^{(c)}_{3,2}(E_{1u},c) = -\frac{\sqrt{35}\mathbb{Q}^{(a)}_{4,1}(E_{2g},2)\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{14} - \frac{\sqrt{35}\mathbb{Q}^{(a)}_{4,2}(E_{2g},2)\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}^{(a)}_{4}(A_{1g})\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{7}$

$\boxed{\text{z435}}$ $\mathbb{Q}^{(c)}_{5,1}(E_{1u},1) = \frac{\mathbb{Q}^{(a)}_{4,1}(E_{2g},1)\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{2} - \frac{\mathbb{Q}^{(a)}_{4,2}(E_{2g},1)\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{2}$

$\boxed{\text{z465}}$ $\mathbb{Q}^{(c)}_{5,2}(E_{1u},1) = -\frac{\mathbb{Q}^{(a)}_{4,1}(E_{2g},1)\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{2} - \frac{\mathbb{Q}^{(a)}_{4,2}(E_{2g},1)\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{2}$

$\boxed{\text{z466}}$ $\mathbb{Q}^{(c)}_{5,1}(E_{1u},2) = \frac{\sqrt{14}\mathbb{Q}^{(a)}_{4,1}(E_{2g},2)\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}^{(a)}_{4,2}(E_{2g},2)\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{14} + \frac{\sqrt{70}\mathbb{Q}^{(a)}_{4}(A_{1g})\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{14}$

$\boxed{\text{z467}}$ $\mathbb{Q}^{(c)}_{5,2}(E_{1u},2) = -\frac{\sqrt{14}\mathbb{Q}^{(a)}_{4,1}(E_{2g},2)\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}^{(a)}_{4,2}(E_{2g},2)\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{14} + \frac{\sqrt{70}\mathbb{Q}^{(a)}_{4}(A_{1g})\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{14}$

$\boxed{\text{z491}}$ $\mathbb{Q}^{(c)}_{2,1}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{0}(A_{1g})\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{2}$

$\boxed{\text{z492}}$ $\mathbb{Q}^{(c)}_{2,2}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{0}(A_{1g})\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{2}$

$\boxed{\text{z493}}$ $\mathbb{Q}^{(c)}_{2,1}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,1}(E_{2g})\mathbb{Q}^{(s)}_{0}(A_{1g})}{2}$

$\boxed{\text{z494}}$ $\mathbb{Q}^{(c)}_{2,2}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,2}(E_{2g})\mathbb{Q}^{(s)}_{0}(A_{1g})}{2}$

$\boxed{\text{z495}}$ $\mathbb{Q}^{(c)}_{2,1}(E_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}^{(a)}_{2}(A_{1g})\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{2}$

$\boxed{\text{z496}}$ $\mathbb{Q}^{(c)}_{2,2}(E_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}^{(a)}_{2}(A_{1g})\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{2}$

$\boxed{\text{z497}}$ $\mathbb{Q}^{(c)}_{2,1}(E_{2g},d) = \frac{\sqrt{4970}\mathbb{Q}^{(a)}_{4,1}(E_{2g},1)\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{142} - \frac{\sqrt{4970}\mathbb{Q}^{(a)}_{4,2}(E_{2g},1)\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{142} + \frac{\sqrt{142}\mathbb{Q}^{(a)}_{4}(A_{1g})\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{142}$

$\boxed{\text{z498}}$ $\mathbb{Q}^{(c)}_{2,2}(E_{2g},d) = -\frac{\sqrt{4970}\mathbb{Q}^{(a)}_{4,1}(E_{2g},1)\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{142} - \frac{\sqrt{4970}\mathbb{Q}^{(a)}_{4,2}(E_{2g},1)\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{142} + \frac{\sqrt{142}\mathbb{Q}^{(a)}_{4}(A_{1g})\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{142}$

$\boxed{\text{z499}}$ $\mathbb{Q}^{(c)}_{4,1}(E_{2g},1a) = \frac{\mathbb{Q}^{(a)}_{2,1}(E_{2g})\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{2} - \frac{\mathbb{Q}^{(a)}_{2,2}(E_{2g})\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{2}$

$\boxed{\text{z500}}$ $\mathbb{Q}^{(c)}_{4,2}(E_{2g},1a) = -\frac{\mathbb{Q}^{(a)}_{2,1}(E_{2g})\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{2} - \frac{\mathbb{Q}^{(a)}_{2,2}(E_{2g})\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{2}$

$\boxed{\text{z501}}$ $\mathbb{Q}^{(c)}_{4,1}(E_{2g},1b) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{4,1}(E_{2g},1)\mathbb{Q}^{(s)}_{0}(A_{1g})}{2}$

$\boxed{\text{z502}}$ $\mathbb{Q}^{(c)}_{4,2}(E_{2g},1b) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{4,2}(E_{2g},1)\mathbb{Q}^{(s)}_{0}(A_{1g})}{2}$

$\boxed{\text{z503}}$ $\mathbb{Q}^{(c)}_{4,1}(E_{2g},1c) = -\frac{\mathbb{Q}^{(a)}_{4,1}(E_{2g},2)\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{2} + \frac{\mathbb{Q}^{(a)}_{4,2}(E_{2g},2)\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{2}$

$\boxed{\text{z504}}$ $\mathbb{Q}^{(c)}_{4,2}(E_{2g},1c) = \frac{\mathbb{Q}^{(a)}_{4,1}(E_{2g},2)\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{2} + \frac{\mathbb{Q}^{(a)}_{4,2}(E_{2g},2)\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{2}$

$\boxed{\text{z505}}$ $\mathbb{Q}^{(c)}_{4,1}(E_{2g},2a) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{4,1}(E_{2g},2)\mathbb{Q}^{(s)}_{0}(A_{1g})}{2}$

$\boxed{\text{z506}}$ $\mathbb{Q}^{(c)}_{4,2}(E_{2g},2a) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{4,2}(E_{2g},2)\mathbb{Q}^{(s)}_{0}(A_{1g})}{2}$

$\boxed{\text{z507}}$ $\mathbb{Q}^{(c)}_{4,1}(E_{2g},2b) = \frac{\sqrt{71}\mathbb{Q}^{(a)}_{4,1}(E_{2g},1)\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{142} - \frac{\sqrt{71}\mathbb{Q}^{(a)}_{4,2}(E_{2g},1)\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{142} - \frac{\sqrt{2485}\mathbb{Q}^{(a)}_{4}(A_{1g})\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{71}$

$\boxed{\text{z508}}$ $\mathbb{Q}^{(c)}_{4,2}(E_{2g},2b) = -\frac{\sqrt{71}\mathbb{Q}^{(a)}_{4,1}(E_{2g},1)\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{142} - \frac{\sqrt{71}\mathbb{Q}^{(a)}_{4,2}(E_{2g},1)\mathbb{Q}^{(s)}_{2,1}(E_{2g})}{142} - \frac{\sqrt{2485}\mathbb{Q}^{(a)}_{4}(A_{1g})\mathbb{Q}^{(s)}_{2,2}(E_{2g})}{71}$

$\boxed{\text{z613}}$ $\mathbb{Q}^{(c)}_{3,1}(E_{2u},a) = -\frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,2}(E_{1g})\mathbb{Q}^{(s)}_{3}(B_{1u})}{2}$

$\boxed{\text{z614}}$ $\mathbb{Q}^{(c)}_{3,2}(E_{2u},a) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,1}(E_{1g})\mathbb{Q}^{(s)}_{3}(B_{1u})}{2}$

$\boxed{\text{z615}}$ $\mathbb{Q}^{(c)}_{3,1}(E_{2u},b) = \frac{\mathbb{Q}^{(a)}_{2,1}(E_{1g})\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{2} - \frac{\mathbb{Q}^{(a)}_{2,2}(E_{1g})\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{2}$

$\boxed{\text{z616}}$ $\mathbb{Q}^{(c)}_{3,2}(E_{2u},b) = -\frac{\mathbb{Q}^{(a)}_{2,1}(E_{1g})\mathbb{Q}^{(s)}_{1,2}(E_{1u})}{2} - \frac{\mathbb{Q}^{(a)}_{2,2}(E_{1g})\mathbb{Q}^{(s)}_{1,1}(E_{1u})}{2}$

$\boxed{\text{z617}}$ $\mathbb{Q}^{(c)}_{3,1}(E_{2u},c) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{4,2}(E_{1g})\mathbb{Q}^{(s)}_{3}(B_{1u})}{2}$

$\boxed{\text{z618}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(s)}(B_{1u})}{2}$

$\boxed{\text{z619}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},d) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8}$

$\boxed{\text{z620}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},d) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8}$

$\boxed{\text{z621}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z622}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2}$

$\boxed{\text{z623}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},2) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8}$

$\boxed{\text{z624}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},2) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{8}$

- `Sn` : '`Sn`' site-cluster
  * bra: $\langle d_v|$ , $\langle d_{xy}|$ , $\langle d_{xz}|$ , $\langle d_{yz}|$ , $\langle d_u|$
  * ket: $|d_v\rangle$ , $|d_{xy}\rangle$ , $|d_{xz}\rangle$ , $|d_{yz}\rangle$ , $|d_u\rangle$
  * wyckoff: `2c`

$\boxed{\text{z7}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g}) = \mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})$

$\boxed{\text{z8}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g}) = \mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})$

$\boxed{\text{z9}}$ $\mathbb{Q}_{4}^{(c)}(A_{1g}) = \mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})$

$\boxed{\text{z70}}$ $\mathbb{Q}_{7}^{(c)}(A_{1u}) = \mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{3}^{(s)}(B_{1u})$

$\boxed{\text{z98}}$ $\mathbb{Q}_{1}^{(c)}(A_{2u}) = \mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{3}^{(s)}(B_{1u})$

$\boxed{\text{z138}}$ $\mathbb{Q}_{4}^{(c)}(B_{1g}) = \mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{0}^{(s)}(A_{1g})$

$\boxed{\text{z139}}$ $\mathbb{Q}_{3}^{(c)}(B_{1u},a) = \mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{3}^{(s)}(B_{1u})$

$\boxed{\text{z140}}$ $\mathbb{Q}_3^{(c)}(B_{1u},b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$

$\boxed{\text{z141}}$ $\mathbb{Q}_3^{(c)}(B_{1u},c) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$

$\boxed{\text{z262}}$ $\mathbb{Q}_4^{(c)}(B_{2g}) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z263}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z264}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z265}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z266}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z267}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},a) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$

$\boxed{\text{z379}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$

$\boxed{\text{z407}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},b) = \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{58}$

$\boxed{\text{z436}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},b) = -\frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{58}$

$\boxed{\text{z437}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{1u}) = \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{58} - \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{29}$

$\boxed{\text{z438}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{1u}) = -\frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_3^{(s)}(B_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_3^{(s)}(B_{1u})}{29}$

$\boxed{\text{z509}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z510}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z511}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z512}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z513}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z514}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z625}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},a) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(s)}(B_{1u})}{2}$

$\boxed{\text{z626}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(s)}(B_{1u})}{2}$

$\boxed{\text{z627}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},b) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(s)}(B_{1u})}{2}$

$\boxed{\text{z628}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},b) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(s)}(B_{1u})}{2}$

- `Mn;Mn_001_1` : 'Mn'-'Mn' bond-cluster
  * bra: $\langle d_v|$ , $\langle d_{xy}|$ , $\langle d_{xz}|$ , $\langle d_{yz}|$ , $\langle d_u|$
  * ket: $|d_v\rangle$ , $|d_{xy}\rangle$ , $|d_{xz}\rangle$ , $|d_{yz}\rangle$ , $|d_u\rangle$
  * wyckoff: `6b@6h`

  $\boxed{\text{z10}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},a) = \mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})$

  $\boxed{\text{z11}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

  $\boxed{\text{z12}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},c) = \mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{M}_{1}^{(b)}(A_{2g})$

$\boxed{\text{z13}}$ $\mathbb{Q}_2^{(c)}(A_{1g},a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z14}}$ $\mathbb{Q}_2^{(c)}(A_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z15}}$ $\mathbb{Q}_2^{(c)}(A_{1g},c) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{M}_1^{(b)}(A_{2g})$

$\boxed{\text{z16}}$ $\mathbb{Q}_2^{(c)}(A_{1g},d) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z17}}$ $\mathbb{Q}_4^{(c)}(A_{1g}) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z18}}$ $\mathbb{Q}_6^{(c)}(A_{1g},2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z44}}$ $\mathbb{Q}_7^{(c)}(A_{1u}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z45}}$ $\mathbb{G}_0^{(c)}(A_{1u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z46}}$ $\mathbb{G}_0^{(c)}(A_{1u},b) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_3^{(b)}(B_{2u})$

$\boxed{\text{z47}}$ $\mathbb{G}_2^{(c)}(A_{1u},a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z71}}$ $\mathbb{G}_2^{(c)}(A_{1u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z72}}$ $\mathbb{G}_4^{(c)}(A_{1u}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z73}}$ $\mathbb{Q}_6^{(c)}(A_{2g}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z74}}$ $\mathbb{G}_1^{(c)}(A_{2g},a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z75}}$ $\mathbb{G}_1^{(c)}(A_{2g},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z76}}$ $\mathbb{G}_3^{(c)}(A_{2g}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z99}}$ $\mathbb{Q}_1^{(c)}(A_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z100}}$ $\mathbb{Q}_1^{(c)}(A_{2u},b) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z101}}$ $\mathbb{Q}_1^{(c)}(A_{2u},c) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z102}}$ $\mathbb{Q}_1^{(c)}(A_{2u},d) = -\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_3^{(b)}(B_{2u})$

$\boxed{\text{z103}}$ $\mathbb{Q}_3^{(c)}(A_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z104}}$ $\mathbb{Q}_3^{(c)}(A_{2u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z142}}$ $\mathbb{Q}_4^{(c)}(B_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z143}}$ $\mathbb{Q}_4^{(c)}(B_{1g},b) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z144}}$ $\mathbb{Q}_4^{(c)}(B_{1g},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z145}}$ $\mathbb{Q}_4^{(c)}(B_{1g},d) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_1^{(b)}(A_{2g})$

$\boxed{\text{z146}}$ $\mathbb{Q}_4^{(c)}(B_{1g},e) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z147}}$ $\mathbb{G}_3^{(c)}(B_{1g}) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z148}}$ $\mathbb{Q}_3^{(c)}(B_{1u},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z149}}$ $\mathbb{Q}_3^{(c)}(B_{1u},b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z150}}$ $\mathbb{Q}_3^{(c)}(B_{1u},c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z151}}$ $\mathbb{Q}_3^{(c)}(B_{1u},d) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z152}}$ $\mathbb{Q}_3^{(c)}(B_{1u},e) = -\frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58}$

$\boxed{\text{z153}}$ $\mathbb{Q}_3^{(c)}(B_{1u},f) = \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u})$

$\boxed{\text{z154}}$ $\mathbb{Q}_3^{(c)}(B_{1u},g) = -\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u})$

$\boxed{\text{z155}}$ $\mathbb{Q}_3^{(c)}(B_{1u},h) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z156}}$ $\mathbb{Q}_5^{(c)}(B_{1u}) = -\frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58} - \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29}$

$\boxed{\text{z157}}$ $\mathbb{Q}_4^{(c)}(B_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z158}}$ $\mathbb{Q}_4^{(c)}(B_{2g},b) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z159}}$ $\mathbb{Q}_4^{(c)}(B_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z160}}$ $\mathbb{Q}_4^{(c)}(B_{2g},d) = \mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_1^{(b)}(A_{2g})$

$\boxed{\text{z161}}$ $\mathbb{Q}_4^{(c)}(B_{2g},e) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z162}}$ $\mathbb{G}_3^{(c)}(B_{2g}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z163}}$ $\mathbb{Q}_3^{(c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z164}}$ $\mathbb{Q}_3^{(c)}(B_{2u},b) = \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58}$

$\boxed{\text{z165}}$ $\mathbb{Q}_{3}^{(c)}(B_{2u},c) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z268}}$ $\mathbb{Q}_{5}^{(c)}(B_{2u}) = \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29}$

$\boxed{\text{z269}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z270}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z271}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z272}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z273}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},c) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8}$

$\boxed{\text{z274}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8}$

$\boxed{\text{z275}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},d) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z276}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},d) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z277}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},e) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z278}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},e) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z279}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},f) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z280}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},f) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z281}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{1g},g) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8} - \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z282}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{1g},g) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z283}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z284}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z285}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},b) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z286}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},b) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z287}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},c) = -\frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8} - \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z288}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},c) = \frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z289}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_{1g},1) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z290}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_{1g},1) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z291}}$ $$\mathbb{G}_{5,1}^{(c)}(E_{1g},1) = \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z292}}$ $$\mathbb{G}_{5,2}^{(c)}(E_{1g},1) = -\frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z293}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z380}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z381}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},b) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z382}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z383}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},c) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z384}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},c) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z385}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},d) = \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58}$

$\boxed{\text{z408}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},d) = -\frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58}$

$\boxed{\text{z409}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},e) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z410}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},e) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z411}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},f) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z412}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},f) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z413}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{1u},a) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{7}$

$\boxed{\text{z439}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{1u},a) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{7}$

$\boxed{\text{z440}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{1u},b) = \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58} - \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29}$

$\boxed{\text{z441}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{1u},b) = -\frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29}$

$\boxed{\text{z442}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},c) = \frac{\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{7}$$

$\boxed{\text{z443}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},c) = -\frac{\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{7}$$

$\boxed{\text{z444}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},d) = -\frac{\sqrt{55}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{22} + \frac{\sqrt{55}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{22} - \frac{\sqrt{33}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{11}$$

$\boxed{\text{z445}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},d) = \frac{\sqrt{55}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{22} + \frac{\sqrt{55}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{22} + \frac{\sqrt{33}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{11}$$

$\boxed{\text{z446}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{1u},1) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z447}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_{1u},1) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z468}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{1u},2) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} + \frac{\sqrt{70}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14}$$

$\boxed{\text{z469}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_{1u},2) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} + \frac{\sqrt{70}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14}$$

$\boxed{\text{z470}}$ $$\mathbb{G}_{2,1}^{(c)}(E_{1u}) = \frac{\sqrt{66}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{22} - \frac{\sqrt{66}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{22} - \frac{\sqrt{110}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{22}$$

$\boxed{\text{z471}}$ $$\mathbb{G}_{2,2}^{(c)}(E_{1u}) = -\frac{\sqrt{66}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{22} - \frac{\sqrt{66}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{22} + \frac{\sqrt{110}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{22}$$

$\boxed{\text{z515}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z516}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z517}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z518}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z519}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z520}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z521}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},d) = \frac{\sqrt{4970}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} - \frac{\sqrt{4970}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} + \frac{\sqrt{142}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142}$

$\boxed{\text{z522}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},d) = -\frac{\sqrt{4970}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} - \frac{\sqrt{4970}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} + \frac{\sqrt{142}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142}$

$\boxed{\text{z523}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},e) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z524}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},e) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z525}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},f) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z526}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},f) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z527}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},g) = -\frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z528}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},g) = \frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z529}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z530}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z531}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z532}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1b) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z533}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1c) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z534}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1c) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z535}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1d) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z536}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1d) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z537}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},2a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z538}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},2a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z539}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},2b) = \frac{\sqrt{71}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} - \frac{\sqrt{71}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} - \frac{\sqrt{2485}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{71}$

$\boxed{\text{z540}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},2b) = -\frac{\sqrt{71}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} - \frac{\sqrt{71}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} - \frac{\sqrt{2485}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{71}$

$\boxed{\text{z629}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},a) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z630}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z631}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z632}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z633}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},c) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z634}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z635}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{2u},d) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8}$$

$\boxed{\text{z636}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{2u},d) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8}$$

$\boxed{\text{z637}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{2u},e) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$$

$\boxed{\text{z638}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{2u},e) = -\frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$$

$\boxed{\text{z639}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{2u},f) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$$

$\boxed{\text{z640}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{2u},f) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$$

$\boxed{\text{z641}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{2u},g) = \frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8} - \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8}$$

$\boxed{\text{z642}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{2u},g) = -\frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8} - \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8} - \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8}$$

$\boxed{\text{z643}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{2u},1) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z644}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_{2u},1) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z645}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{2u},2) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8}$$

$\boxed{\text{z646}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_{2u},2) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8}$$

$\boxed{\text{z647}}$ $$\mathbb{G}_{2,1}^{(c)}(E_{2u},a) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z648}}$ $$\mathbb{G}_{2,2}^{(c)}(E_{2u},a) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z649}}$ $\mathbb{G}_{2,1}^{(c)}(E_{2u},b) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8}$

$\boxed{\text{z650}}$ $\mathbb{G}_{2,2}^{(c)}(E_{2u},b) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8} - \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{8}$

$\boxed{\text{z651}}$ $\mathbb{G}_{4,1}^{(c)}(E_{2u},1) = -\frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z652}}$ $\mathbb{G}_{4,2}^{(c)}(E_{2u},1) = \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

- `Sn;Mn_001_1` : 'Mn'-'Sn' bond-cluster
  * bra: $\langle d_v|$ , $\langle d_{xy}|$ , $\langle d_{xz}|$ , $\langle d_{yz}|$ , $\langle d_u|$
  * ket: $|d_v\rangle$ , $|d_{xy}\rangle$ , $|d_{xz}\rangle$ , $|d_{yz}\rangle$ , $|d_u\rangle$
  * wyckoff: `12a@12j`

$\boxed{\text{z19}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},a) = \mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})$

$\boxed{\text{z20}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z21}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},c) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z22}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},d) = \mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{M}_{1}^{(b)}(A_{2g})$

$\boxed{\text{z23}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},a) = \mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})$

$\boxed{\text{z24}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z25}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},c) = \mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{M}_{1}^{(b)}(A_{2g})$

$\boxed{\text{z26}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},d) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z27}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},e) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z28}}$ $\mathbb{Q}_4^{(c)}(A_{1g}) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z29}}$ $\mathbb{Q}_6^{(c)}(A_{1g}, 2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2}$

$\boxed{\text{z30}}$ $\mathbb{Q}_6^{(c)}(A_{1g}, 2b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z31}}$ $\mathbb{Q}_6^{(c)}(A_{1g}, 2c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2}$

$\boxed{\text{z48}}$ $\mathbb{Q}_7^{(c)}(A_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2}$

$\boxed{\text{z49}}$ $\mathbb{Q}_7^{(c)}(A_{1u}, b) = \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2} + \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_3^{(b)}(B_{2u})}{2}$

$\boxed{\text{z50}}$ $\mathbb{Q}_7^{(c)}(A_{1u}, c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2}$

$\boxed{\text{z51}}$ $\mathbb{G}_0^{(c)}(A_{1u}, a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a)}{2}$

$\boxed{\text{z52}}$ $\mathbb{G}_0^{(c)}(A_{1u}, b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b)}{2}$

$\boxed{\text{z53}}$ $\mathbb{G}_0^{(c)}(A_{1u}, c) = \frac{\sqrt{2}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} + \frac{\sqrt{2}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_3^{(b)}(B_{2u})}{2}$

$\boxed{\text{z54}}$ $\mathbb{G}_2^{(c)}(A_{1u}, a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z55}}$ $\mathbb{G}_2^{(c)}(A_{1u}, b) = \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_3^{(b)}(B_{2u})}{2}$

$\boxed{\text{z56}}$ $\mathbb{G}_2^{(c)}(A_{1u}, c) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a)}{2}$

$\boxed{\text{z57}}$ $\mathbb{G}_2^{(c)}(A_{1u}, d) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b)}{2}$

$\boxed{\text{z58}}$ $\mathbb{G}_4^{(c)}(A_{1u}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z59}}$ $\mathbb{G}_6^{(c)}(A_{1u}, 2) = -\frac{\sqrt{2}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} + \frac{\sqrt{2}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_3^{(b)}(B_{2u})}{2}$

$\boxed{\text{z60}}$ $\mathbb{Q}_6^{(c)}(A_{2g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_6^{(b)}(A_{2g})$

$\boxed{\text{z77}}$ $\mathbb{Q}_6^{(c)}(A_{2g}, b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_6^{(b)}(A_{2g})$

$\boxed{\text{z78}}$ $\mathbb{Q}_6^{(c)}(A_{2g}, c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2}$

$\boxed{\text{z79}}$ $\mathbb{Q}_6^{(c)}(A_{2g}, d) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_6^{(b)}(A_{2g})$

$\boxed{\text{z80}}$ $\mathbb{Q}_6^{(c)}(A_{2g}, e) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z81}}$ $\mathbb{Q}_6^{(c)}(A_{2g}, f) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2}$

$\boxed{\text{z82}}$ $\mathbb{G}_1^{(c)}(A_{2g}, a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z83}}$ $\mathbb{G}_1^{(c)}(A_{2g}, b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2}$

$\boxed{\text{z84}}$ $\mathbb{G}_1^{(c)}(A_{2g}, c) = \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_0^{(b)}(A_{1g})$

$\boxed{\text{z85}}$ $\mathbb{G}_1^{(c)}(A_{2g}, d) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a)}{2}$

$\boxed{\text{z86}}$ $\mathbb{G}_1^{(c)}(A_{2g}, e) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b)}{2}$

$\boxed{\text{z87}}$ $\mathbb{G}_3^{(c)}(A_{2g}, a) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z88}}$ $\mathbb{G}_3^{(c)}(A_{2g}, b) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_0^{(b)}(A_{1g})$

$\boxed{\text{z105}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z106}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},b) = \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z107}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},c) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2}$$

$\boxed{\text{z108}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},d) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2}$$

$\boxed{\text{z109}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},e) = -\frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} + \frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z110}}$ $$\mathbb{Q}_3^{(c)}(A_{2u},a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z111}}$ $$\mathbb{Q}_3^{(c)}(A_{2u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2}$$

$\boxed{\text{z112}}$ $$\mathbb{Q}_3^{(c)}(A_{2u},c) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2}$$

$\boxed{\text{z113}}$ $$\mathbb{Q}_7^{(c)}(A_{2u},2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z114}}$ $$\mathbb{Q}_7^{(c)}(A_{2u},2b) = \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z115}}$ $$\mathbb{Q}_7^{(c)}(A_{2u},2c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z116}}$ $$\mathbb{G}_6^{(c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} + \frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z166}}$ $$\mathbb{Q}_4^{(c)}(B_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z167}}$ $$\mathbb{Q}_4^{(c)}(B_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$$

$\boxed{\text{z168}}$ $\mathbb{Q}_4^{(c)}(B_{1g},c) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z169}}$ $\mathbb{Q}_4^{(c)}(B_{1g},d) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_6^{(b)}(A_{2g})$

$\boxed{\text{z170}}$ $\mathbb{Q}_4^{(c)}(B_{1g},e) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z171}}$ $\mathbb{Q}_4^{(c)}(B_{1g},f) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z172}}$ $\mathbb{Q}_4^{(c)}(B_{1g},g) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_1^{(b)}(A_{2g})$

$\boxed{\text{z173}}$ $\mathbb{Q}_4^{(c)}(B_{1g},h) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z174}}$ $\mathbb{Q}_4^{(c)}(B_{1g},i) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z175}}$ $\mathbb{G}_3^{(c)}(B_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z176}}$ $\mathbb{G}_3^{(c)}(B_{1g},b) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z177}}$ $\mathbb{G}_3^{(c)}(B_{1g},c) = \mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_0^{(b)}(A_{1g})$

$\boxed{\text{z178}}$ $\mathbb{Q}_3^{(c)}(B_{1u},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z179}}$ $\mathbb{Q}_3^{(c)}(B_{1u},b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z180}}$ $\mathbb{Q}_3^{(c)}(B_{1u},c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z181}}$ $\mathbb{Q}_3^{(c)}(B_{1u},d) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z182}}$ $\mathbb{Q}_3^{(c)}(B_{1u},e) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$

$\boxed{\text{z183}}$ $\mathbb{Q}_3^{(c)}(B_{1u},f) = -\frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58}$

$\boxed{\text{z184}}$ $\mathbb{Q}_3^{(c)}(B_{1u},g) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z185}}$ $\mathbb{Q}_3^{(c)}(B_{1u},h) = \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u})$

$\boxed{\text{z186}}$ $\mathbb{Q}_3^{(c)}(B_{1u},i) = -\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u})$

$\boxed{\text{z187}}$ $\mathbb{Q}_3^{(c)}(B_{1u},j) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2}$

$\boxed{\text{z188}}$ $\mathbb{Q}_3^{(c)}(B_{1u},k) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2}$

$\boxed{\text{z189}}$ $\mathbb{Q}_5^{(c)}(B_{1u}) = -\frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58} - \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29}$

$\boxed{\text{z190}}$ $\mathbb{Q}_9^{(c)}(B_{1u},1) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z191}}$ $\mathbb{Q}_4^{(c)}(B_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z192}}$ $\mathbb{Q}_4^{(c)}(B_{2g},b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z193}}$ $\mathbb{Q}_4^{(c)}(B_{2g},c) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z194}}$ $\mathbb{Q}_4^{(c)}(B_{2g},d) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_6^{(b)}(A_{2g})$

$\boxed{\text{z195}}$ $\mathbb{Q}_4^{(c)}(B_{2g},e) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z196}}$ $\mathbb{Q}_4^{(c)}(B_{2g},f) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z197}}$ $\mathbb{Q}_4^{(c)}(B_{2g},g) = \mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_1^{(b)}(A_{2g})$

$\boxed{\text{z198}}$ $\mathbb{Q}_4^{(c)}(B_{2g},h) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z199}}$ $\mathbb{Q}_4^{(c)}(B_{2g},i) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z200}}$ $\mathbb{G}_3^{(c)}(B_{2g},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z201}}$ $\mathbb{G}_3^{(c)}(B_{2g},b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z202}}$ $\mathbb{G}_3^{(c)}(B_{2g},c) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_0^{(b)}(A_{1g})$

$\boxed{\text{z203}}$ $\mathbb{Q}_3^{(c)}(B_{2u},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{2u})$

$\boxed{\text{z204}}$ $\mathbb{Q}_3^{(c)}(B_{2u},b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{2u})$

$\boxed{\text{z205}}$ $\mathbb{Q}_3^{(c)}(B_{2u},c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z206}}$ $\mathbb{Q}_3^{(c)}(B_{2u},d) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z207}}$ $\mathbb{Q}_3^{(c)}(B_{2u},e) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{2u})$

$\boxed{\text{z208}}$ $\mathbb{Q}_3^{(c)}(B_{2u},f) = \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29} - \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58}$

$\boxed{\text{z209}}$ $\mathbb{Q}_3^{(c)}(B_{2u},g) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z210}}$ $\mathbb{Q}_3^{(c)}(B_{2u},h) = -\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{1u})$

$\boxed{\text{z211}}$ $\mathbb{Q}_3^{(c)}(B_{2u},i) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{1u})$

$\boxed{\text{z212}}$ $\mathbb{Q}_3^{(c)}(B_{2u},j) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2}$

$\boxed{\text{z213}}$ $\mathbb{Q}_{3}^{(c)}(B_{2u},k) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2}$

$\boxed{\text{z294}}$ $\mathbb{Q}_{5}^{(c)}(B_{2u}) = \frac{\sqrt{58}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{29} + \frac{\sqrt{58}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{58} + \frac{\sqrt{406}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{29}$

$\boxed{\text{z295}}$ $\mathbb{Q}_{9}^{(c)}(B_{2u},1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z296}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z297}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z298}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z299}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z300}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},c) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8}$

$\boxed{\text{z301}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8}$

$\boxed{\text{z302}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},d) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z303}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},d) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z304}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},e) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z305}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},e) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z306}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},f) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z307}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{1g},f) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2}$$

$\boxed{\text{z308}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{1g},g) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$$

$\boxed{\text{z309}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{1g},g) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2}$$

$\boxed{\text{z310}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{1g},h) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$$

$\boxed{\text{z311}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{1g},h) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$$

$\boxed{\text{z312}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{1g},i) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8} - \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8}$$

$\boxed{\text{z313}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{1g},i) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8}$$

$\boxed{\text{z314}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{1g},j) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8} - \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8}$$

$\boxed{\text{z315}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{1g},j) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8}$$

$\boxed{\text{z316}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z317}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z318}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},b) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z319}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},b) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{8}$$

$\boxed{\text{z320}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},c) = -\frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8} - \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8}$$

$\boxed{\text{z321}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{1g},c) = \frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{8}$

$\boxed{\text{z322}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{1g},d) = -\frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8} - \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8}$

$\boxed{\text{z323}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{1g},d) = \frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{8}$

$\boxed{\text{z324}}$ $\mathbb{Q}_{6,1}^{(c)}(E_{1g},1a) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{6}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z325}}$ $\mathbb{Q}_{6,2}^{(c)}(E_{1g},1a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{6}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z326}}$ $\mathbb{Q}_{6,1}^{(c)}(E_{1g},1b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z327}}$ $\mathbb{Q}_{6,2}^{(c)}(E_{1g},1b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z328}}$ $\mathbb{Q}_{6,1}^{(c)}(E_{1g},1c) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{6}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z329}}$ $\mathbb{Q}_{6,2}^{(c)}(E_{1g},1c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{6}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z330}}$ $\mathbb{Q}_{6,1}^{(c)}(E_{1g},1d) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z331}}$ $\mathbb{Q}_{6,2}^{(c)}(E_{1g},1d) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z332}}$ $\mathbb{Q}_{6,1}^{(c)}(E_{1g},1e) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z333}}$ $\mathbb{Q}_{6,2}^{(c)}(E_{1g},1e) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z334}}$ $\mathbb{Q}_{8,1}^{(c)}(E_{1g},1) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z335}}$ $\mathbb{Q}_{8,2}^{(c)}(E_{1g},1) = -\frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} + \frac{\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z336}}$ $\mathbb{G}_{1,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z337}}$ $\mathbb{G}_{1,2}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z338}}$ $\mathbb{G}_{3,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z339}}$ $\mathbb{G}_{3,2}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z340}}$ $\mathbb{G}_{5,1}^{(c)}(E_{1g},1a) = \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2} + \frac{\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z341}}$ $\mathbb{G}_{5,2}^{(c)}(E_{1g},1a) = -\frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2} + \frac{\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z342}}$ $\mathbb{G}_{5,1}^{(c)}(E_{1g},1b) = \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2} + \frac{\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z343}}$ $\mathbb{G}_{5,2}^{(c)}(E_{1g},1b) = -\frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2} + \frac{\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z344}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z345}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z386}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{2u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$

$\boxed{\text{z387}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{2u})}{2}$

$\boxed{\text{z388}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},c) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z389}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},c) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14}$$

$\boxed{\text{z390}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},d) = \frac{\sqrt{203}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{2u})}{29} - \frac{\sqrt{29}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{2u})}{58} + \frac{\sqrt{203}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29} + \frac{\sqrt{29}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58}$$

$\boxed{\text{z391}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},d) = -\frac{\sqrt{203}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29} - \frac{\sqrt{29}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58} + \frac{\sqrt{203}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{2u})}{29} - \frac{\sqrt{29}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{2u})}{58}$$

$\boxed{\text{z392}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},e) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z393}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},e) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z394}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},f) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2}$$

$\boxed{\text{z395}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},f) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2}$$

$\boxed{\text{z396}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},g) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2}$$

$\boxed{\text{z397}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},g) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2}$$

$\boxed{\text{z414}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},h) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z415}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},h) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$$

$\boxed{\text{z416}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},a) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{7}$$

$\boxed{\text{z417}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},a) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{7}$$

$\boxed{\text{z418}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},b) = \frac{\sqrt{29}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{2u})}{58} + \frac{\sqrt{203}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{2u})}{29} + \frac{\sqrt{29}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58} - \frac{\sqrt{203}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29}$$

$\boxed{\text{z419}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},b) = -\frac{\sqrt{29}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{58} + \frac{\sqrt{203}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{29} + \frac{\sqrt{29}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{2u})}{58} + \frac{\sqrt{203}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{2u})}{29}$$

$\boxed{\text{z420}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},c) = \frac{\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{7}$$

$\boxed{\text{z421}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},c) = -\frac{\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{7}$$

$\boxed{\text{z422}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},d) = -\frac{\sqrt{55}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{22} + \frac{\sqrt{55}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{22} - \frac{\sqrt{33}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{11}$$

$\boxed{\text{z423}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},d) = \frac{\sqrt{55}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{22} + \frac{\sqrt{55}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{22} + \frac{\sqrt{33}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{11}$$

$\boxed{\text{z424}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{1u},e) = -\frac{\sqrt{55}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{22} + \frac{\sqrt{55}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{22} - \frac{\sqrt{33}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{11}$$

$\boxed{\text{z425}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_{1u},e) = \frac{\sqrt{55}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{22} + \frac{\sqrt{55}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{22} + \frac{\sqrt{33}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{11}$$

$\boxed{\text{z448}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{1u},1a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z449}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_{1u},1a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z450}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{1u},1b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z451}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_{1u},1b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2}$$

$\boxed{\text{z452}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{1u},1c) = -\frac{\sqrt{2}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z453}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_{1u},1c) = -\frac{\sqrt{2}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$$

$\boxed{\text{z454}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_{1u},1d) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z455}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{1u},1d) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{1u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z456}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{1u},1e) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z457}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{1u},1e) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z458}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{1u},1f) = \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z459}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{1u},1f) = \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z460}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{1u},1g) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z472}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{1u},1g) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2} - \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z473}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{1u},2) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} + \frac{\sqrt{70}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z474}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{1u},2) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{14} + \frac{\sqrt{70}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z475}}$ $\mathbb{Q}_{7,1}^{(c)}(E_{1u},1a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z476}}$ $\mathbb{Q}_{7,2}^{(c)}(E_{1u},1a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z477}}$ $\mathbb{Q}_{7,1}^{(c)}(E_{1u},1b) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z478}}$ $\mathbb{Q}_{7,2}^{(c)}(E_{1u},1b) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{1u})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z479}}$ $\mathbb{Q}_{7,1}^{(c)}(E_{1u},1c) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z480}}$ $\mathbb{Q}_{7,2}^{(c)}(E_{1u},1c) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z481}}$ $\mathbb{G}_{2,1}^{(c)}(E_{1u},a) = \frac{\sqrt{66}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{22} - \frac{\sqrt{66}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{22} - \frac{\sqrt{110}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{22}$

$\boxed{\text{z482}}$ $\mathbb{G}_{2,2}^{(c)}(E_{1u},a) = -\frac{\sqrt{66}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{22} - \frac{\sqrt{66}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{22} + \frac{\sqrt{110}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{22}$

$\boxed{\text{z483}}$ $\mathbb{G}_{2,1}^{(c)}(E_{1u},b) = \frac{\sqrt{66}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{22} - \frac{\sqrt{66}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{22} - \frac{\sqrt{110}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{22}$

$\boxed{\text{z484}}$ $\mathbb{G}_{2,2}^{(c)}(E_{1u},b) = -\frac{\sqrt{66}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{22} - \frac{\sqrt{66}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{22} + \frac{\sqrt{110}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{22}$

$\boxed{\text{z541}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z542}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z543}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z544}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z545}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z546}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},c) = -\frac{\sqrt{2}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z547}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},d) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z548}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},d) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z549}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},e) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{6}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z550}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},e) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{6}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z551}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},f) = \frac{\sqrt{4970}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} - \frac{\sqrt{4970}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} + \frac{\sqrt{142}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142}$

$\boxed{\text{z552}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},f) = -\frac{\sqrt{4970}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} - \frac{\sqrt{4970}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} + \frac{\sqrt{142}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142}$

$\boxed{\text{z553}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},g) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z554}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},g) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z555}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},h) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z556}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},h) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z557}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},i) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z558}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},i) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z559}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},j) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z560}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},j) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{1}^{(b)}(A_{2g})}{2}$

$\boxed{\text{z561}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},k) = -\frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z562}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},k) = \frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z563}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{2g},l) = -\frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z564}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{2g},l) = \frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z565}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z566}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z567}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1b) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_6^{(b)}(A_{2g})}{2}$

$\boxed{\text{z568}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_6^{(b)}(A_{2g})}{2}$

$\boxed{\text{z569}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z570}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z571}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1d) = -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z572}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1d) = -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z573}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1e) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z574}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1e) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z575}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1f) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_6^{(b)}(A_{2g})}{2}$

$\boxed{\text{z576}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1f) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_6^{(b)}(A_{2g})}{2}$

$\boxed{\text{z577}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1g) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z578}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1g) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z579}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1h) = \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z580}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1h) = \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z581}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1i) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z582}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1i) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},a)}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},a)}{2}$

$\boxed{\text{z583}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},1j) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z584}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},1j) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g},b)}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g},b)}{2}$

$\boxed{\text{z585}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},2a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z586}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},2a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z587}}$ $\mathbb{Q}_{4,1}^{(c)}(E_{2g},2b) = \frac{\sqrt{71}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} - \frac{\sqrt{71}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} - \frac{\sqrt{2485}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{71}$

$\boxed{\text{z588}}$ $\mathbb{Q}_{4,2}^{(c)}(E_{2g},2b) = -\frac{\sqrt{71}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{142} - \frac{\sqrt{71}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{142} - \frac{\sqrt{2485}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{71}$

$\boxed{\text{z589}}$ $\mathbb{Q}_{8,1}^{(c)}(E_{2g},1) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z590}}$ $\mathbb{Q}_{8,2}^{(c)}(E_{2g},1) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{4,2}^{(b)}(E_{2g},1)}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{4,1}^{(b)}(E_{2g},1)}{2}$

$\boxed{\text{z591}}$ $\mathbb{G}_{3,1}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z592}}$ $\mathbb{G}_{3,2}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z653}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z654}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z655}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z656}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z657}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},c) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z658}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},c) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z659}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},d) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8}$

$\boxed{\text{z660}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},d) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8}$

$\boxed{\text{z661}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},e) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z662}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},e) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z663}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},f) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z664}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},f) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z665}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},g) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z666}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},g) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z667}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},h) = \frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8} - \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8}$

$\boxed{\text{z668}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},h) = -\frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8} - \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8} - \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8}$

$\boxed{\text{z669}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},i) = \frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8} - \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8}$

$\boxed{\text{z670}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},i) = -\frac{\sqrt{10}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8} - \frac{\sqrt{10}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8} - \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8}$

$\boxed{\text{z671}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z672}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z673}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z674}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z675}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1c) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z676}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1c) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{1u})}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z677}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1d) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z678}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1d) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z679}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1e) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z680}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1e) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z681}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1f) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z682}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1f) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2} - \frac{\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z683}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},2) = \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8}$

$\boxed{\text{z684}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},2) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{8} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{8}$

$\boxed{\text{z685}}$ $\mathbb{Q}_{9,1}^{(c)}(E_{2u},1) = -\frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z686}}$ $\mathbb{Q}_{9,2}^{(c)}(E_{2u},1) = \frac{\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u},1)}{2} + \frac{\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u},1)}{2}$

$\boxed{\text{z687}}$ $\mathbb{G}_{2,1}^{(c)}(E_{2u},a) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2}$

$\boxed{\text{z688}}$ $\mathbb{G}_{2,2}^{(c)}(E_{2u},a) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2}$

$\boxed{\text{z689}}$ $\mathbb{G}_{2,1}^{(c)}(E_{2u},b) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2}$

$\boxed{\text{z690}}$ $\mathbb{G}_{2,2}^{(c)}(E_{2u},b) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2}$

$\boxed{\text{z691}}$ $\mathbb{G}_{2,1}^{(c)}(E_{2u},c) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8}$

$\boxed{\text{z692}}$ $\mathbb{G}_{2,2}^{(c)}(E_{2u},c) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8} - \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{8}$

$\boxed{\text{z693}}$ $\mathbb{G}_{2,1}^{(c)}(E_{2u},d) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8}$

$\boxed{\text{z694}}$ $\mathbb{G}_{2,2}^{(c)}(E_{2u},d) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8} - \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{8} + \frac{\sqrt{10}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{8}$

$\boxed{\text{z695}}$ $\mathbb{G}_{4,1}^{(c)}(E_{2u},1a) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z696}}$ $\mathbb{G}_{4,2}^{(c)}(E_{2u},1a) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{2u})}{2}$

$\boxed{\text{z697}}$ $\mathbb{G}_{4,1}^{(c)}(E_{2u},1b) = -\frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2}$

$\boxed{\text{z698}}$ $\mathbb{G}_{4,2}^{(c)}(E_{2u},1b) = \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},a)}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},a)}{2}$

$\boxed{\text{z699}}$ $\mathbb{G}_{4,1}^{(c)}(E_{2u},1c) = -\frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2}$

$\boxed{\text{z700}}$ $\mathbb{G}_{4,2}^{(c)}(E_{2u},1c) = \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u},b)}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u},b)}{2}$

- `Sn;Sn_001_1` : 'Sn'-'Sn' bond-cluster
  * bra: $\langle d_v|$ , $\langle d_{xy}|$ , $\langle d_{xz}|$ , $\langle d_{yz}|$ , $\langle d_u|$
  * ket: $|d_v\rangle$ , $|d_{xy}\rangle$ , $|d_{xz}\rangle$ , $|d_{yz}\rangle$ , $|d_u\rangle$
  * wyckoff: `6a@6g`

$\boxed{\text{z32}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},a) = \mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})$

$\boxed{\text{z33}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2} + \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z34}}$ $\mathbb{Q}_{0}^{(c)}(A_{1g},c) = \mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{4}^{(b)}(B_{2g})$

$\boxed{\text{z35}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},a) = \mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})$

$\boxed{\text{z36}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2} - \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z37}}$ $\mathbb{Q}_{2}^{(c)}(A_{1g},c) = \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{3} + \frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6} + \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{3} + \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6}$

$\boxed{\text{z38}}$ $\mathbb{Q}_4^{(c)}(A_{1g},a) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z39}}$ $\mathbb{Q}_4^{(c)}(A_{1g},b) = \frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{3} + \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{3}$

$\boxed{\text{z40}}$ $\mathbb{Q}_6^{(c)}(A_{1g},2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z61}}$ $\mathbb{G}_0^{(c)}(A_{1u},a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{3}$

$\boxed{\text{z62}}$ $\mathbb{G}_0^{(c)}(A_{1u},b) = \mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_3^{(b)}(B_{1u})$

$\boxed{\text{z63}}$ $\mathbb{G}_2^{(c)}(A_{1u},a) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{3}$

$\boxed{\text{z64}}$ $\mathbb{G}_2^{(c)}(A_{1u},b) = \frac{\sqrt{14}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{7} + \frac{\sqrt{14}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{7} + \frac{\sqrt{21}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{7}$

$\boxed{\text{z65}}$ $\mathbb{G}_2^{(c)}(A_{1u},c) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z66}}$ $\mathbb{G}_4^{(c)}(A_{1u}) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{14} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{14} + \frac{2\sqrt{7}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{7}$

$\boxed{\text{z89}}$ $\mathbb{Q}_6^{(c)}(A_{2g},a) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_4^{(b)}(B_{2g})$

$\boxed{\text{z90}}$ $\mathbb{Q}_6^{(c)}(A_{2g},b) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$

$\boxed{\text{z91}}$ $\mathbb{G}_1^{(c)}(A_{2g}) = \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{10} - \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{5} - \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{5}$

$\boxed{\text{z92}}$ $\mathbb{G}_3^{(c)}(A_{2g},a) = \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{5} + \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{10} - \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{5} - \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{10}$

$\boxed{\text{z93}}$ $\mathbb{G}_3^{(c)}(A_{2g},b) = \frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{3} - \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} + \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{3}$

$\boxed{\text{z94}}$ $$\mathbb{G}_5^{(c)}(A_{2g}) = \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{3} + \frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{3} - \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6}$$

$\boxed{\text{z117}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z118}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},b) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_3^{(b)}(B_{1u})$$

$\boxed{\text{z119}}$ $$\mathbb{Q}_1^{(c)}(A_{2u},c) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$$

$\boxed{\text{z120}}$ $$\mathbb{Q}_3^{(c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z121}}$ $$\mathbb{Q}_4^{(c)}(B_{1g},a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2}$$

$\boxed{\text{z122}}$ $$\mathbb{Q}_4^{(c)}(B_{1g},b) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z123}}$ $$\begin{aligned}\mathbb{Q}_4^{(c)}(B_{1g},c) = &-\frac{3\sqrt{71}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{71} - \frac{\sqrt{994}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{71} - \frac{5\sqrt{142}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{142}\\ &+ \frac{3\sqrt{71}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{71} + \frac{\sqrt{994}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{71} + \frac{5\sqrt{142}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{142}\end{aligned}$$

$\boxed{\text{z124}}$ $$\begin{aligned}\mathbb{Q}_6^{(c)}(B_{1g}) = &\frac{7\sqrt{497}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{355} + \frac{9\sqrt{142}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{710} - \frac{6\sqrt{994}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{355}\\ &- \frac{7\sqrt{497}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{355} - \frac{9\sqrt{142}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{710} + \frac{6\sqrt{994}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{355}\end{aligned}$$

$\boxed{\text{z125}}$ $$\mathbb{G}_3^{(c)}(B_{1g},a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2}$$

$\boxed{\text{z214}}$ $$\mathbb{G}_3^{(c)}(B_{1g},b) = -\frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{10} + \frac{\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{5} - \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{5} + \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{10} - \frac{\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{5} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{5}$$

$\boxed{\text{z215}}$ $$\mathbb{Q}_3^{(c)}(B_{1u},a) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$$

$\boxed{\text{z216}}$ $$\mathbb{Q}_3^{(c)}(B_{1u},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{4} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{4} - \frac{\sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_1^{(b)}(A_{2u})}{2}$$

$\boxed{\text{z217}}$ $$\mathbb{Q}_3^{(c)}(B_{1u},c) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$$

$\boxed{\text{z218}}$ $$\mathbb{G}_4^{(c)}(B_{1u}) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{4} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{4} + \frac{\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_1^{(b)}(A_{2u})}{2}$$

$\boxed{\text{z219}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_4^{(b)}(B_{2g})$$

$\boxed{\text{z220}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_4^{(b)}(B_{2g})$$

$\boxed{\text{z221}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2}$$

$\boxed{\text{z222}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},d) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z223}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},e) = -\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_4^{(b)}(B_{2g})$$

$\boxed{\text{z224}}$ $$\begin{aligned}\mathbb{Q}_4^{(c)}(B_{2g},f) = &-\frac{3\sqrt{71}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{71} - \frac{\sqrt{994}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{71} + \frac{5\sqrt{142}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{142}\\ &- \frac{3\sqrt{71}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{71} - \frac{\sqrt{994}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{71} + \frac{5\sqrt{142}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{142}\end{aligned}$$

$\boxed{\text{z225}}$ $$\begin{aligned}\mathbb{Q}_6^{(c)}(B_{2g}) = &\frac{7\sqrt{497}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{355} + \frac{9\sqrt{142}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{710} + \frac{6\sqrt{994}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{355}\\ &+ \frac{7\sqrt{497}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{355} + \frac{9\sqrt{142}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{710} + \frac{6\sqrt{994}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{355}\end{aligned}$$

$\boxed{\text{z226}}$ $$\mathbb{G}_3^{(c)}(B_{2g},a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2}$$

$\boxed{\text{z227}}$ $$\mathbb{G}_3^{(c)}(B_{2g},b) = \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{10} - \frac{\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{5} - \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{5} + \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{10} - \frac{\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{5} - \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{5}$$

$\boxed{\text{z228}}$ $$\mathbb{Q}_3^{(c)}(B_{2u},a) = -\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{1u})$$

$\boxed{\text{z229}}$ $$\mathbb{Q}_3^{(c)}(B_{2u},b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$$

$\boxed{\text{z230}}$ $\mathbb{Q}_3^{(c)}(B_{2u},c) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{4} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{4} + \frac{\sqrt{3}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{2}$

$\boxed{\text{z231}}$ $\mathbb{Q}_3^{(c)}(B_{2u},d) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{1u})$

$\boxed{\text{z232}}$ $\mathbb{Q}_3^{(c)}(B_{2u},e) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z233}}$ $\mathbb{G}_4^{(c)}(B_{2u}) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{4} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{4} + \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{2}$

$\boxed{\text{z234}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2}$

$\boxed{\text{z235}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2}$

$\boxed{\text{z236}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z237}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z238}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_4^{(b)}(B_{2g})}{2}$

$\boxed{\text{z239}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},c) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_4^{(b)}(B_{2g})}{2}$

$\boxed{\text{z240}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},d) = -\frac{\sqrt{78}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{26} - \frac{\sqrt{78}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{26} + \frac{\sqrt{78}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{26} + \frac{\sqrt{78}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{26} + \frac{\sqrt{26}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{26}$

$\boxed{\text{z241}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},d) = \frac{\sqrt{78}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{26} + \frac{\sqrt{78}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{26} + \frac{\sqrt{78}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{26} + \frac{\sqrt{78}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{26} + \frac{\sqrt{26}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{26}$

$\boxed{\text{z242}}$ $\mathbb{Q}_{2,1}^{(c)}(E_{1g},e) = -\frac{\sqrt{742}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_4^{(b)}(B_{2g})}{53} + \frac{5\sqrt{106}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_4^{(b)}(B_{2g})}{106}$

$\boxed{\text{z243}}$ $\mathbb{Q}_{2,2}^{(c)}(E_{1g},e) = -\frac{\sqrt{742}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_4^{(b)}(B_{2g})}{53} + \frac{5\sqrt{106}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_4^{(b)}(B_{2g})}{106}$

$\boxed{\text{z346}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{1g},f) = \frac{\sqrt{30}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{48} - \frac{\sqrt{15}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{12} - \frac{\sqrt{30}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{48} + \frac{\sqrt{15}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{12} - \frac{\sqrt{3}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} + \frac{\sqrt{210}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{48} + \frac{\sqrt{210}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{48}$$

$\boxed{\text{z347}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{1g},f) = -\frac{\sqrt{30}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{48} + \frac{\sqrt{15}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{12} - \frac{\sqrt{30}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{48} + \frac{\sqrt{15}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{12} - \frac{\sqrt{3}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{210}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{48} + \frac{\sqrt{210}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{48}$$

$\boxed{\text{z348}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},a) = \frac{\sqrt{26}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{52} + \frac{\sqrt{26}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{52} - \frac{\sqrt{26}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{52} - \frac{\sqrt{26}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{52} + \frac{\sqrt{78}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{13}$$

$\boxed{\text{z349}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},a) = -\frac{\sqrt{26}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{52} - \frac{\sqrt{26}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{52} - \frac{\sqrt{26}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{52} - \frac{\sqrt{26}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{52} + \frac{\sqrt{78}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{13}$$

$\boxed{\text{z350}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z351}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},b) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z352}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},c) = \frac{5\sqrt{106}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{4}^{(b)}(B_{2g})}{106} + \frac{\sqrt{742}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{4}^{(b)}(B_{2g})}{53}$$

$\boxed{\text{z353}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},c) = \frac{5\sqrt{106}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{4}^{(b)}(B_{2g})}{106} + \frac{\sqrt{742}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{4}^{(b)}(B_{2g})}{53}$$

$\boxed{\text{z354}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{1g},d) = -\frac{125\sqrt{2274}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{18192} - \frac{43\sqrt{1137}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{4548} + \frac{125\sqrt{2274}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{18192} + \frac{43\sqrt{1137}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{4548} + \frac{\sqrt{5685}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2274} - \frac{29\sqrt{15918}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{18192} - \frac{29\sqrt{15918}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{18192}$$

$\boxed{\text{z355}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{1g},d) = \frac{125\sqrt{2274}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{18192} + \frac{43\sqrt{1137}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{4548} + \frac{125\sqrt{2274}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{18192} + \frac{43\sqrt{1137}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{4548} + \frac{\sqrt{5685}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2274} + \frac{29\sqrt{15918}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{18192} - \frac{29\sqrt{15918}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{18192}$$

$\boxed{\text{z356}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_{1g},1) = \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{6}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6} + \frac{\sqrt{6}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6}$$

$\boxed{\text{z357}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_{1g},1) = -\frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} + \frac{\sqrt{6}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6} + \frac{\sqrt{6}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6}$$

$\boxed{\text{z358}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_{1g},2) = \frac{\sqrt{2653}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{758} - \frac{2\sqrt{5306}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{1137} - \frac{\sqrt{2653}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{758} + \frac{2\sqrt{5306}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{1137} + \frac{3\sqrt{26530}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{758} + \frac{17\sqrt{379}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2274} + \frac{17\sqrt{379}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2274}$$

$\boxed{\text{z359}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_{1g},2) = -\frac{\sqrt{2653}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{758} + \frac{2\sqrt{5306}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{1137} - \frac{\sqrt{2653}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{758} + \frac{2\sqrt{5306}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{1137} + \frac{3\sqrt{26530}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{758} - \frac{17\sqrt{379}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2274} + \frac{17\sqrt{379}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2274}$$

$\boxed{\text{z360}}$ $$\mathbb{G}_{1,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{4} - \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{4} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{4} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{4}$$

$\boxed{\text{z361}}$ $$\mathbb{G}_{1,2}^{(c)}(E_{1g}) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{4} + \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{4} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{4} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{4}$$

$\boxed{\text{z362}}$ $$\mathbb{G}_{3,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{4} - \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{4} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{12} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{12}$$

$\boxed{\text{z363}}$ $$\mathbb{G}_{3,2}^{(c)}(E_{1g}) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{4} + \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{4} + \frac{\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} + \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{12} - \frac{\sqrt{14}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{12}$$

$\boxed{\text{z364}}$ $$\mathbb{G}_{5,1}^{(c)}(E_{1g},1) = \frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} - \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} + \frac{\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6}$$

$\boxed{\text{z365}}$ $$\mathbb{G}_{5,2}^{(c)}(E_{1g},1) = -\frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6} - \frac{\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6}$$

$\boxed{\text{z366}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},a) = \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{2} - \frac{\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z367}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_{1u},a) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{2} + \frac{\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$$

$\boxed{\text{z368}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_{1u},b) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$$

$\boxed{\text{z369}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},b) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z370}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},c) = -\frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z371}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},c) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z372}}$ $\mathbb{Q}_{1,1}^{(c)}(E_{1u},d) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} - \frac{\sqrt{15}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} + \frac{\sqrt{15}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8}$

$\boxed{\text{z373}}$ $\mathbb{Q}_{1,2}^{(c)}(E_{1u},d) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} + \frac{\sqrt{15}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} + \frac{\sqrt{15}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8}$

$\boxed{\text{z374}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{1u},a) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{12} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{12} - \frac{\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z375}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{1u},a) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{12} + \frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{12} + \frac{\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z398}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{1u},b) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} - \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8}$

$\boxed{\text{z399}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{1u},b) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} + \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8}$

$\boxed{\text{z400}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{1u},1) = \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z401}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{1u},1) = -\frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z402}}$ $\mathbb{G}_{2,1}^{(c)}(E_{1u},a) = \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{2} + \frac{\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z403}}$ $\mathbb{G}_{2,2}^{(c)}(E_{1u},a) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{2} - \frac{\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z426}}$ $\mathbb{G}_{2,1}^{(c)}(E_{1u},b) = \frac{\sqrt{210}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{42} + \frac{2\sqrt{21}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{21} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{42} - \frac{\sqrt{14}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z427}}$ $$\mathbb{G}_{2,2}^{(c)}(E_{1u},b) = -\frac{2\sqrt{21}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{21} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{42} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{42} + \frac{\sqrt{14}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{14}$$

$\boxed{\text{z428}}$ $$\mathbb{G}_{4,1}^{(c)}(E_{1u}) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{28} + \frac{\sqrt{210}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{28} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{28} + \frac{\sqrt{35}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{14}$$

$\boxed{\text{z429}}$ $$\mathbb{G}_{4,2}^{(c)}(E_{1u}) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1}^{(b)}(A_{2u})}{28} + \frac{\sqrt{21}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{28} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{28} - \frac{\sqrt{35}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{14}$$

$\boxed{\text{z461}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z462}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z463}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z464}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z485}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4}^{(b)}(B_{2g})}{2}$$

$\boxed{\text{z486}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},c) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4}^{(b)}(B_{2g})}{2}$$

$\boxed{\text{z487}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},d) = -\frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{10} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{10} - \frac{\sqrt{5}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{5}$$

$\boxed{\text{z488}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},d) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{10} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{10} - \frac{\sqrt{5}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{5}$$

$\boxed{\text{z489}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},e) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{4}^{(b)}(B_{2g})}{2}$$

$\boxed{\text{z490}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},e) = -\frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{4}^{(b)}(B_{2g})}{2}$$

$\boxed{\text{z593}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_{2g},f) = \frac{\sqrt{555}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{222} + \frac{\sqrt{7770}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{222} - \frac{\sqrt{555}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{222} - \frac{\sqrt{7770}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{222} + \frac{\sqrt{222}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{222} - \frac{\sqrt{3885}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{222} + \frac{\sqrt{3885}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{222}$$

$\boxed{\text{z594}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_{2g},f) = -\frac{\sqrt{555}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{222} - \frac{\sqrt{7770}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{222} - \frac{\sqrt{555}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{222} - \frac{\sqrt{7770}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{222} + \frac{\sqrt{222}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{222} + \frac{\sqrt{3885}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{222} + \frac{\sqrt{3885}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{222}$$

$\boxed{\text{z595}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{2g},1a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z596}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{2g},1a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$\boxed{\text{z597}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{2g},1b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z598}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{2g},1b) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z599}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{2g},1c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6} - \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6}$$

$\boxed{\text{z600}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{2g},1c) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6} + \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6} + \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} - \frac{\sqrt{7}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6}$$

$\boxed{\text{z601}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{2g},2a) = -\frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{10} + \frac{\sqrt{30}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{10}$$

$\boxed{\text{z602}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{2g},2a) = \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{10} + \frac{\sqrt{30}\mathbb{Q}_{2}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{10}$$

$\boxed{\text{z603}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{2g},2b) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z604}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{2g},2b) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z605}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_{2g},2c) = -\frac{137\sqrt{30747}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{122988} - \frac{13\sqrt{430458}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{61494} + \frac{137\sqrt{30747}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{122988} + \frac{13\sqrt{430458}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{61494} - \frac{47\sqrt{307470}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{61494} - \frac{85\sqrt{215229}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{122988} + \frac{85\sqrt{215229}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{122988}$$

$\boxed{\text{z606}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_{2g},2c) = \frac{137\sqrt{30747}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{122988} + \frac{13\sqrt{430458}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{61494} + \frac{137\sqrt{30747}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{122988} + \frac{13\sqrt{430458}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{61494} - \frac{47\sqrt{307470}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{61494} + \frac{85\sqrt{215229}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{122988} + \frac{85\sqrt{215229}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{122988}$$

$\boxed{\text{z607}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_{2g},1) = \frac{\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6} - \frac{\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6}$$

$\boxed{\text{z608}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_{2g},1) = -\frac{\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g},2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{6} - \frac{\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g},2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{6} + \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} - \frac{\sqrt{2}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6}$$

$\boxed{\text{z609}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_{2g},2) = -\frac{17\sqrt{11634}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{4155} + \frac{11\sqrt{831}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2770} + \frac{17\sqrt{11634}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{4155} - \frac{11\sqrt{831}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2770} + \frac{7\sqrt{29085}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{4155} + \frac{\sqrt{1662}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{1385} - \frac{\sqrt{1662}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{1385}$$

$\boxed{\text{z610}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_{2g},2) = \frac{17\sqrt{11634}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{4155} - \frac{11\sqrt{831}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2770} + \frac{17\sqrt{11634}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{4155} - \frac{11\sqrt{831}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2770} + \frac{7\sqrt{29085}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{4155} - \frac{\sqrt{1662}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{1385} - \frac{\sqrt{1662}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{1385}$$

$\boxed{\text{z611}}$ $$\mathbb{G}_{3,1}^{(c)}(E_{2g}) = -\frac{\sqrt{21}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{60} + \frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{10} + \frac{\sqrt{21}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{60} - \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{10} - \frac{\sqrt{210}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{30} + \frac{3\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{20} - \frac{3\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{20}$$

$\boxed{\text{z612}}$ $$\mathbb{G}_{3,2}^{(c)}(E_{2g}) = \frac{\sqrt{21}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{60} - \frac{\sqrt{6}\mathbb{Q}_{4,1}^{(a)}(E_{2g},1)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{10} + \frac{\sqrt{21}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{60} - \frac{\sqrt{6}\mathbb{Q}_{4,2}^{(a)}(E_{2g},1)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{10} - \frac{\sqrt{210}\mathbb{Q}_{4}^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{30} - \frac{3\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{20} - \frac{3\sqrt{3}\mathbb{Q}_{4}^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{20}$$

$\boxed{\text{z701}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_{2u},a) = -\frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$$

$\boxed{\text{z702}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z703}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},b) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z704}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},b) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z705}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},c) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{12} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{12} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{6} + \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{4} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{4}$

$\boxed{\text{z706}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},c) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{12} + \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{6} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{12} - \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{4} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{4}$

$\boxed{\text{z707}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},d) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z708}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},d) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{3}^{(b)}(B_{1u})}{2}$

$\boxed{\text{z709}}$ $\mathbb{Q}_{3,1}^{(c)}(E_{2u},e) = \frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z710}}$ $\mathbb{Q}_{3,2}^{(c)}(E_{2u},e) = -\frac{\sqrt{2}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z711}}$ $\mathbb{Q}_{5,1}^{(c)}(E_{2u},1) = -\frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z712}}$ $\mathbb{Q}_{5,2}^{(c)}(E_{2u},1) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} + \frac{\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2}$

$\boxed{\text{z713}}$ $\mathbb{G}_{2,1}^{(c)}(E_{2u},a) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z714}}$ $\mathbb{G}_{2,2}^{(c)}(E_{2u},a) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z715}}$ $\mathbb{G}_{2,1}^{(c)}(E_{2u},b) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{42} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{42} + \frac{\sqrt{210}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{42} + \frac{\sqrt{35}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{14} + \frac{\sqrt{35}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z716}}$ $\mathbb{G}_{2,2}^{(c)}(E_{2u},b) = \frac{\sqrt{21}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{42} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{42} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{42} - \frac{\sqrt{35}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{14} + \frac{\sqrt{35}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{14}$

$\boxed{\text{z717}}$ $\mathbb{G}_{4,1}^{(c)}(E_{2u},1) = -\frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z718}}$ $\mathbb{G}_{4,2}^{(c)}(E_{2u},1) = \frac{\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} + \frac{\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$

$\boxed{\text{z719}}$ $\mathbb{G}_{4,1}^{(c)}(E_{2u},2) = \frac{\sqrt{105}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{14} - \frac{\sqrt{7}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{28} - \frac{\sqrt{7}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{28}$

$\boxed{\text{z720}}$ $\mathbb{G}_{4,2}^{(c)}(E_{2u},2) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{28} - \frac{\sqrt{42}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1}^{(b)}(A_{2u})}{14} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{28} - \frac{\sqrt{7}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{28}$

## Atomic SAMB

- bra: $\langle d_v|$ , $\langle d_{xy}|$ , $\langle d_{xz}|$ , $\langle d_{yz}|$ , $\langle d_u|$
- ket: $|d_v\rangle$ , $|d_{xy}\rangle$ , $|d_{xz}\rangle$ , $|d_{yz}\rangle$ , $|d_u\rangle$

$\boxed{\text{x1}}$ $\mathbb{Q}_{0}^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$

$\boxed{\text{x2}}$ $\mathbb{Q}_{2}^{(a)}(A_{1g}) = \begin{bmatrix} -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \end{bmatrix}$

$\boxed{\text{x3}}$ $$\mathbb{Q}_4^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{35} \end{bmatrix}$$

$\boxed{\text{x4}}$ $$\mathbb{Q}_4^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x5}}$ $$\mathbb{Q}_4^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x6}}$ $$\mathbb{Q}_{2,1}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \end{bmatrix}$$

$\boxed{\text{x7}}$ $$\mathbb{Q}_{2,2}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x8}}$ $$\mathbb{Q}^{(a)}_{4,1}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{\sqrt{21}}{7} \\ 0 & 0 & 0 & \frac{\sqrt{21}}{7} & 0 \end{bmatrix}$$

$\boxed{\text{x9}}$ $$\mathbb{Q}^{(a)}_{4,2}(E_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}}{7} \\ 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{7} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x10}}$ $$\mathbb{Q}^{(a)}_{2,1}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x11}}$ $$\mathbb{Q}^{(a)}_{2,2}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x12}}$ $$\mathbb{Q}^{(a)}_{4,1}(E_{2g}, 1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x13}}$ $\mathbb{Q}_{4,2}^{(a)}(E_{2g},1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x14}}$ $\mathbb{Q}_{4,1}^{(a)}(E_{2g},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{7} & 0 \\ \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x15}}$ $\mathbb{Q}_{4,2}^{(a)}(E_{2g},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{7} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x16}}$ $\mathbb{M}_{1}^{(a)}(A_{2g}) = \begin{bmatrix} 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x17}}$ $\mathbb{M}_{3}^{(a)}(A_{2g}) = \begin{bmatrix} 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x18}}$ $\mathbb{M}_3^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x19}}$ $\mathbb{M}_3^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x20}}$ $\mathbb{M}_{1,1}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{10} & 0 \end{bmatrix}$

$\boxed{\text{x21}}$ $\mathbb{M}_{1,2}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & \frac{\sqrt{30}i}{10} \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{10} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x22}}$ $\mathbb{M}_{3,1}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 \end{bmatrix}$

$\boxed{\text{x23}}$ $\mathbb{M}_{3,2}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x24}}$ $\mathbb{M}_{3,1}^{(a)}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x25}}$ $\mathbb{M}_{3,2}^{(a)}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$

## Cluster SAMB

- Site cluster

** Wyckoff: `2c`

$\boxed{\text{y1}}$ $\mathbb{Q}_{0}^{(s)}(A_{1g}) = \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$

$\boxed{\text{y2}}$ $\mathbb{Q}_{3}^{(s)}(B_{1u}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

** Wyckoff: `6h`

$\boxed{\text{y3}}$ $\mathbb{Q}_{0}^{(s)}(A_{1g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y4}}$ $\mathbb{Q}_3^{(s)}(B_{1u}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y5}}$ $\mathbb{Q}_{1,1}^{(s)}(E_{1u}) = \left[0, -\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}\right]$

$\boxed{\text{y6}}$ $\mathbb{Q}_{1,2}^{(s)}(E_{1u}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y7}}$ $\mathbb{Q}_{2,1}^{(s)}(E_{2g}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y8}}$ $\mathbb{Q}_{2,2}^{(s)}(E_{2g}) = \left[0, \frac{1}{2}, -\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}\right]$

- Bond cluster

** Wyckoff: `6b@6h`

$\boxed{\text{y9}}$ $\mathbb{Q}_0^{(s)}(A_{1g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y10}}$ $\mathbb{M}_1^{(s)}(A_{2g}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}\right]$

$\boxed{\text{y11}}$ $\mathbb{Q}_3^{(s)}(B_{1u}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y12}}$ $\mathbb{T}_3^{(s)}(B_{2u}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}\right]$

$\boxed{\text{y13}}$ $\mathbb{Q}_{1,1}^{(s)}(E_{1u}) = \left[0, -\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}\right]$

$\boxed{\text{y14}}$ $\mathbb{Q}_{1,2}^{(s)}(E_{1u}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y15}}$ $\mathbb{T}_{1,1}^{(s)}(E_{1u}) = \left[\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}\right]$

$\boxed{\text{y16}}$ $\mathbb{T}^{(s)}_{1,2}(E_{1u}) = \left[0,\ \frac{i}{2},\ -\frac{i}{2},\ 0,\ -\frac{i}{2},\ \frac{i}{2}\right]$

$\boxed{\text{y17}}$ $\mathbb{Q}^{(s)}_{2,1}(E_{2g}) = \left[\frac{\sqrt{3}}{3},\ -\frac{\sqrt{3}}{6},\ -\frac{\sqrt{3}}{6},\ \frac{\sqrt{3}}{3},\ -\frac{\sqrt{3}}{6},\ -\frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y18}}$ $\mathbb{Q}^{(s)}_{2,2}(E_{2g}) = \left[0,\ \frac{1}{2},\ -\frac{1}{2},\ 0,\ \frac{1}{2},\ -\frac{1}{2}\right]$

$\boxed{\text{y19}}$ $\mathbb{T}^{(s)}_{2,1}(E_{2g}) = \left[0,\ \frac{i}{2},\ -\frac{i}{2},\ 0,\ \frac{i}{2},\ -\frac{i}{2}\right]$

$\boxed{\text{y20}}$ $\mathbb{T}^{(s)}_{2,2}(E_{2g}) = \left[-\frac{\sqrt{3}i}{3},\ \frac{\sqrt{3}i}{6},\ \frac{\sqrt{3}i}{6},\ -\frac{\sqrt{3}i}{3},\ \frac{\sqrt{3}i}{6},\ \frac{\sqrt{3}i}{6}\right]$

** Wyckoff: `6a@6g`

$\boxed{\text{y21}}$ $\mathbb{Q}^{(s)}_{0}(A_{1g}) = \left[\frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y22}}$ $\mathbb{T}^{(s)}_{1}(A_{2u}) = \left[\frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6}\right]$

$\boxed{\text{y23}}$ $\mathbb{T}^{(s)}_{3}(B_{1u}) = \left[\frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ -\frac{\sqrt{6}i}{6},\ -\frac{\sqrt{6}i}{6},\ -\frac{\sqrt{6}i}{6}\right]$

$\boxed{\text{y24}}$ $\mathbb{Q}^{(s)}_{4}(B_{2g}) = \left[\frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ -\frac{\sqrt{6}}{6},\ -\frac{\sqrt{6}}{6},\ -\frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y25}}$ $\mathbb{Q}^{(s)}_{2,1}(E_{1g}) = \left[\frac{\sqrt{3}}{3},\ -\frac{\sqrt{3}}{6},\ -\frac{\sqrt{3}}{6},\ -\frac{\sqrt{3}}{3},\ \frac{\sqrt{3}}{6},\ \frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y26}}$ $\mathbb{Q}^{(s)}_{2,2}(E_{1g}) = \left[0,\ \frac{1}{2},\ -\frac{1}{2},\ 0,\ -\frac{1}{2},\ \frac{1}{2}\right]$

$\boxed{\text{y27}}$ $\mathbb{T}^{(s)}_{1,1}(E_{1u}) = \left[0,\ -\frac{i}{2},\ \frac{i}{2},\ 0,\ \frac{i}{2},\ -\frac{i}{2}\right]$

$\boxed{\text{y28}}$ $\mathbb{T}^{(s)}_{1,2}(E_{1u}) = \left[\frac{\sqrt{3}i}{3},\ -\frac{\sqrt{3}i}{6},\ -\frac{\sqrt{3}i}{6},\ -\frac{\sqrt{3}i}{3},\ \frac{\sqrt{3}i}{6},\ \frac{\sqrt{3}i}{6}\right]$

$\boxed{\text{y29}}$ $\mathbb{Q}_{2,1}^{(s)}(E_{2g}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y30}}$ $\mathbb{Q}_{2,2}^{(s)}(E_{2g}) = \left[0, \frac{1}{2}, -\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}\right]$

$\boxed{\text{y31}}$ $\mathbb{M}_{2,1}^{(s)}(E_{2u}) = \left[0, -\frac{i}{2}, \frac{i}{2}, 0, -\frac{i}{2}, \frac{i}{2}\right]$

$\boxed{\text{y32}}$ $\mathbb{M}_{2,2}^{(s)}(E_{2u}) = \left[\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}\right]$

** Wyckoff: `12a@12j`

$\boxed{\text{y33}}$ $\mathbb{Q}_{0}^{(s)}(A_{1g}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y34}}$ $\mathbb{T}_{0}^{(s)}(A_{1g}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}\right]$

$\boxed{\text{y35}}$ $\mathbb{M}_{1}^{(s)}(A_{2g}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}\right]$

$\boxed{\text{y36}}$ $\mathbb{Q}_{6}^{(s)}(A_{2g}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y37}}$ $\mathbb{Q}_{3}^{(s)}(B_{1u}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y38}}$ $\mathbb{T}_{3}^{(s)}(B_{1u}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}\right]$

$\boxed{\text{y39}}$ $\mathbb{Q}_{3}^{(s)}(B_{2u}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y40}}$ $\mathbb{T}_{3}^{(s)}(B_{2u}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}\right]$

$\boxed{\text{y41}}$ $\mathbb{Q}^{(s)}_{1,1}(E_{1u}) = \left[\frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, -\frac{\sqrt{42}}{84}, -\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, -\frac{\sqrt{42}}{84}, \frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, -\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21}\right]$

$\boxed{\text{y42}}$ $\mathbb{Q}^{(s)}_{1,2}(E_{1u}) = \left[\frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14}, -\frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{28}, \frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14}, -\frac{3\sqrt{14}}{28}, \frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14}\right]$

$\boxed{\text{y43}}$ $\mathbb{T}^{(s)}_{1,1}(E_{1u}, a) = \left[\frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, -\frac{\sqrt{42}i}{84}, -\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{84}, \frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, -\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21}\right]$

$\boxed{\text{y44}}$ $\mathbb{T}^{(s)}_{1,2}(E_{1u}, a) = \left[\frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14}, -\frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14}, \frac{3\sqrt{14}i}{28}, \frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14}, -\frac{3\sqrt{14}i}{28}, \frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14}\right]$

$\boxed{\text{y45}}$ $\mathbb{T}^{(s)}_{1,1}(E_{1u}, b) = \left[\frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14}, -\frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14}, \frac{3\sqrt{14}i}{28}, -\frac{3\sqrt{14}i}{28}, \frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14}, \frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14}\right]$

$\boxed{\text{y46}}$ $\mathbb{T}^{(s)}_{1,2}(E_{1u}, b) = \left[-\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, \frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, -\frac{\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{84}, \frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, -\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21}\right]$

$\boxed{\text{y47}}$ $\mathbb{Q}^{(s)}_{5,1}(E_{1u}, 1) = \left[\frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14}, -\frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{28}, -\frac{3\sqrt{14}}{28}, \frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14}\right]$

$\boxed{\text{y48}}$ $\mathbb{Q}^{(s)}_{5,2}(E_{1u}, 1) = \left[-\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, \frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, -\frac{\sqrt{42}}{84}, -\frac{\sqrt{42}}{84}, \frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, -\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21}\right]$

$\boxed{\text{y49}}$ $\mathbb{Q}^{(s)}_{2,1}(E_{2g}) = \left[\frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}\right]$

$\boxed{\text{y50}}$ $\mathbb{Q}^{(s)}_{2,2}(E_{2g}) = \left[-\frac{5\sqrt{2}}{28}, \frac{2\sqrt{2}}{7}, -\frac{3\sqrt{2}}{28}, -\frac{5\sqrt{2}}{28}, \frac{2\sqrt{2}}{7}, -\frac{3\sqrt{2}}{28}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}\right]$

$\boxed{\text{y51}}$ $\mathbb{T}^{(s)}_{2,1}(E_{2g}, a) = \left[\frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}\right]$

$\boxed{\text{y52}}$ $\mathbb{T}^{(s)}_{2,2}(E_{2g}, a) = \left[-\frac{5\sqrt{2}i}{28}, \frac{2\sqrt{2}i}{7}, -\frac{3\sqrt{2}i}{28}, -\frac{5\sqrt{2}i}{28}, \frac{2\sqrt{2}i}{7}, -\frac{3\sqrt{2}i}{28}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}\right]$

$\boxed{\text{y53}}$ $\mathbb{T}^{(s)}_{2,1}(E_{2g}, b) = \left[\frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}\right]$

$\boxed{\text{y54}}$ $\mathbb{T}^{(s)}_{2,2}(E_{2g}, b) = \left[\frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{13\sqrt{6}i}{84}, -\frac{11\sqrt{6}i}{84}, -\frac{\sqrt{6}i}{42}, \frac{13\sqrt{6}i}{84}, -\frac{11\sqrt{6}i}{84}, -\frac{\sqrt{6}i}{42}\right]$

$\boxed{\text{y55}}$ $\mathbb{Q}^{(s)}_{4,1}(E_{2g}, 1) = \left[\frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}\right]$

$\boxed{\text{y56}}$ $\mathbb{Q}^{(s)}_{4,2}(E_{2g}, 1) = \left[\frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{13\sqrt{6}}{84}, -\frac{11\sqrt{6}}{84}, -\frac{\sqrt{6}}{42}, \frac{13\sqrt{6}}{84}, -\frac{11\sqrt{6}}{84}, -\frac{\sqrt{6}}{42}\right]$

## Site and Bond

Table 5: Orbital of each site

| # | site | orbital |
|---|---|---|
| 1 | `Mn` | $\lvert d_v\rangle$, $\lvert d_{xy}\rangle$, $\lvert d_{xz}\rangle$, $\lvert d_{yz}\rangle$, $\lvert d_u\rangle$ |
| 2 | `Sn` | $\lvert d_v\rangle$, $\lvert d_{xy}\rangle$, $\lvert d_{xz}\rangle$, $\lvert d_{yz}\rangle$, $\lvert d_u\rangle$ |

Table 6: Neighbor and bra-ket of each bond

| # | head | tail | neighbor | head (bra) | tail (ket) |
|---|---|---|---|---|---|
| 1 | `Mn` | `Mn` | `[1]` | `[d]` | `[d]` |
| 2 | `Mn` | `Sn` | `[1]` | `[d]` | `[d]` |
| 3 | `Sn` | `Sn` | `[1]` | `[d]` | `[d]` |

## Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'Mn' (#1) site cluster (6h), mm2

| SL | position ($\boldsymbol{s}$) | mapping |
|---|---|---|
| 1 | [ 0.83880, 0.67760, 0.25000] | [1,11,16,20] |
| 2 | [ 0.32240, 0.16120, 0.25000] | [2,10,17,19] |
| 3 | [ 0.83880, 0.16120, 0.25000] | [3,12,18,21] |
| 4 | [ 0.16120, 0.32240, 0.75000] | [4,8,13,23] |
| 5 | [ 0.67760, 0.83880, 0.75000] | [5,7,14,22] |
| 6 | [ 0.16120, 0.83880, 0.75000] | [6,9,15,24] |

Table 8: 'Sn' (#2) site cluster (2c), -6m2

| SL | position ($\boldsymbol{s}$) | mapping |
|---|---|---|
| 1 | [ 0.33333, 0.66667, 0.25000] | [1,2,3,10,11,12,16,17,18,19,20,21] |
| 2 | [ 0.66667, 0.33333, 0.75000] | [4,5,6,7,8,9,13,14,15,22,23,24] |

## Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 9: 1-th 'Mn'-'Mn' [1] (#1) bond cluster (6b@6h), ND, $|\boldsymbol{v}|$= 0.4836 (cartesian)

| SL | vector ($\boldsymbol{v}$) | center ($\boldsymbol{c}$) | mapping | head | tail | $\boldsymbol{R}$ (primitive) |
|---|---|---|---|---|---|---|
| 1 | [-0.48360, 0.00000, 0.00000] | [ 0.08060, 0.16120, 0.25000] | [1,-11,16,-20] | (3,1) | (2,1) | [1,0,0] |
| 2 | [ 0.00000,-0.48360, 0.00000] | [ 0.83880, 0.91940, 0.25000] | [2,-10,17,-19] | (1,1) | (3,1) | [0,1,0] |
| 3 | [ 0.48360, 0.48360, 0.00000] | [ 0.08060, 0.91940, 0.25000] | [3,-12,18,-21] | (2,1) | (1,1) | [-1,-1,0] |
| 4 | [ 0.48360, 0.00000, 0.00000] | [ 0.91940, 0.83880, 0.75000] | [4,-8,13,-23] | (6,1) | (5,1) | [-1,0,0] |
| 5 | [ 0.00000, 0.48360, 0.00000] | [ 0.16120, 0.08060, 0.75000] | [5,-7,14,-22] | (4,1) | (6,1) | [0,-1,0] |
| 6 | [-0.48360,-0.48360, 0.00000] | [ 0.91940, 0.08060, 0.75000] | [6,-9,15,-24] | (5,1) | (4,1) | [1,1,0] |

Table 10: 1-th 'Mn'-'Sn' [1] (#2) bond cluster (12a@12j), D, $|\boldsymbol{v}|$= 0.50009 (cartesian)

| SL | vector ($\boldsymbol{v}$) | center ($\boldsymbol{c}$) | mapping | head | tail | $\boldsymbol{R}$ (primitive) |
|---|---|---|---|---|---|---|
| 1 | [-0.49453, 0.01093, 0.00000] | [ 0.08607, 0.67213, 0.25000] | [1,16] | (1,1) | (1,1) | [1,0,0] |
| 2 | [-0.01093,-0.50547, 0.00000] | [ 0.32787, 0.41393, 0.25000] | [2,17] | (2,1) | (1,1) | [0,0,0] |
| 3 | [ 0.50547, 0.49453, 0.00000] | [ 0.58607, 0.91393, 0.25000] | [3,18] | (3,1) | (1,1) | [0,-1,0] |
| 4 | [ 0.49453,-0.01093, 0.00000] | [ 0.91393, 0.32787, 0.75000] | [4,13] | (4,1) | (2,1) | [-1,0,0] |
| 5 | [ 0.01093, 0.50547, 0.00000] | [ 0.67213, 0.58607, 0.75000] | [5,14] | (5,1) | (2,1) | [0,0,0] |
| 6 | [-0.50547,-0.49453, 0.00000] | [ 0.41393, 0.08607, 0.75000] | [6,15] | (6,1) | (2,1) | [0,1,0] |
| 7 | [ 0.01093,-0.49453, 0.00000] | [ 0.67213, 0.08607, 0.75000] | [7,22] | (5,1) | (2,1) | [0,1,0] |
| 8 | [-0.50547,-0.01093, 0.00000] | [ 0.41393, 0.32787, 0.75000] | [8,23] | (4,1) | (2,1) | [0,0,0] |
| 9 | [ 0.49453, 0.50547, 0.00000] | [ 0.91393, 0.58607, 0.75000] | [9,24] | (6,1) | (2,1) | [-1,0,0] |

*continued ...*

Table 10

| SL | vector ($\boldsymbol{v}$) | center ($\boldsymbol{c}$) | mapping | head | tail | $\boldsymbol{R}$ (primitive) |
|---|---|---|---|---|---|---|
| 10 | [-0.01093, 0.49453, 0.00000] | [ 0.32787, 0.91393, 0.25000] | [10,19] | (2,1) | (1,1) | [0,-1,0] |
| 11 | [ 0.50547, 0.01093, 0.00000] | [ 0.58607, 0.67213, 0.25000] | [11,20] | (1,1) | (1,1) | [0,0,0] |
| 12 | [-0.49453,-0.50547, 0.00000] | [ 0.08607, 0.41393, 0.25000] | [12,21] | (3,1) | (1,1) | [1,0,0] |

Table 11: 1-th 'Sn'-'Sn' [1] (#3) bond cluster (6a@6g), ND, $|\boldsymbol{v}|$= 0.76376 (cartesian)

| SL | vector ($\boldsymbol{v}$) | center ($\boldsymbol{c}$) | mapping | head | tail | $\boldsymbol{R}$ (primitive) |
|---|---|---|---|---|---|---|
| 1 | [ 0.33333, 0.66667,-0.50000] | [ 0.50000, 0.00000, 0.00000] | [1,-8,-13,20] | (2,1) | (1,1) | [0,-1,1] |
| 2 | [-0.66667,-0.33333,-0.50000] | [ 0.00000, 0.50000, 0.00000] | [2,-7,-14,19] | (2,1) | (1,1) | [1,0,1] |
| 3 | [ 0.33333,-0.33333,-0.50000] | [ 0.50000, 0.50000, 0.00000] | [3,-9,-15,21] | (2,1) | (1,1) | [0,0,1] |
| 4 | [-0.33333,-0.66667,-0.50000] | [ 0.50000, 0.00000, 0.50000] | [4,-11,-16,23] | (1,1) | (2,1) | [0,1,0] |
| 5 | [ 0.66667, 0.33333,-0.50000] | [ 0.00000, 0.50000, 0.50000] | [5,-10,-17,22] | (1,1) | (2,1) | [-1,0,0] |
| 6 | [-0.33333, 0.33333,-0.50000] | [ 0.50000, 0.50000, 0.50000] | [6,-12,-18,24] | (1,1) | (2,1) | [0,0,0] |